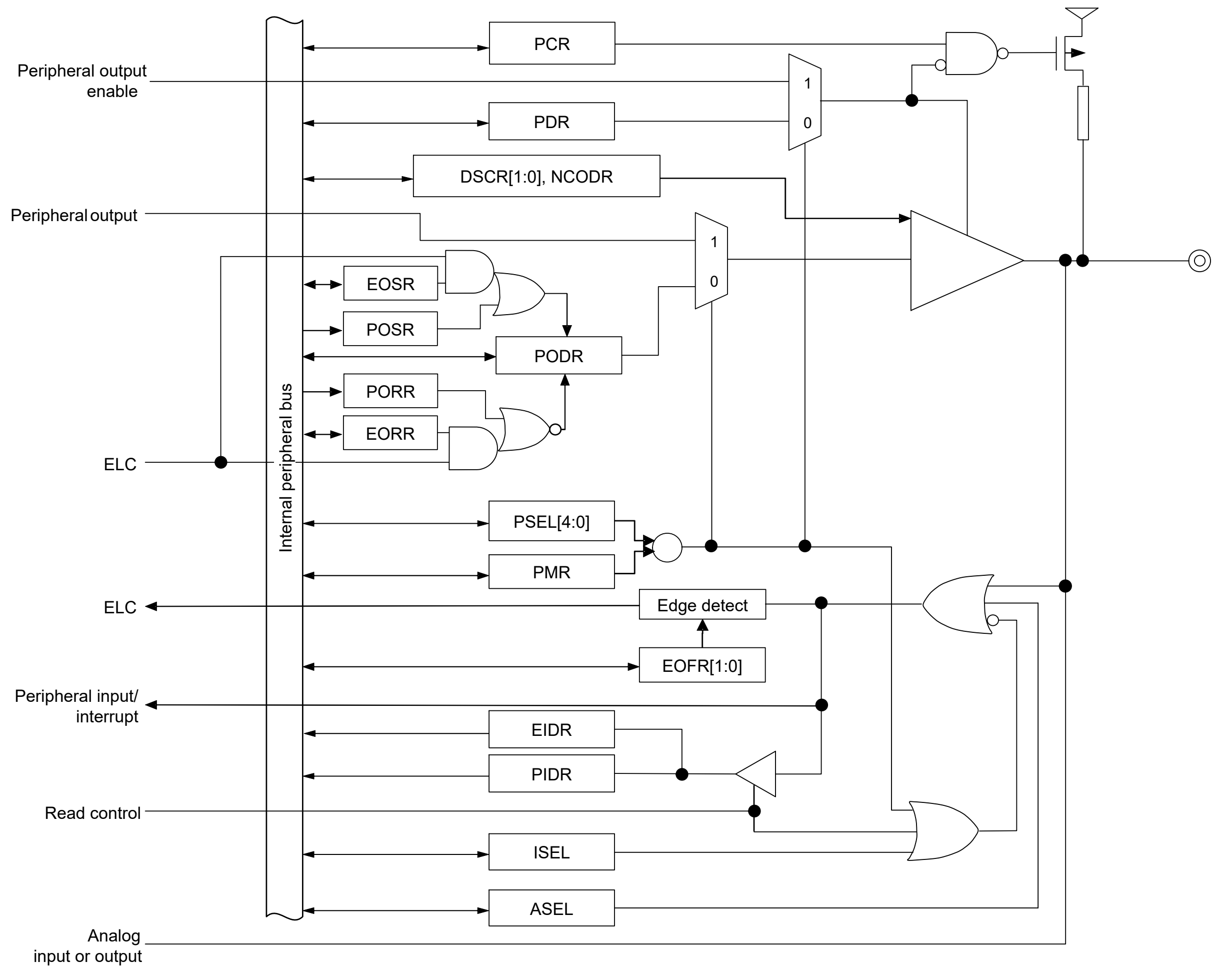
21. I/O Ports

21.1 Overview

The I/O port pins operate as general I/O port pins, I/O pins for peripheral modules, interrupt input pins, analog I/O, port group function for the ELC or bus control pins.

All pins except P209 (as TDO of JTAG ports) operate as input pins immediately after a reset, and pin functions are switched by register settings. The I/O ports and peripheral modules for each pin are specified in the associated registers.

Figure 21.1 shows a connection diagram for the I/O port registers. The configuration of the I/O ports differs for different packages. Table 21.1 lists the I/O port specifications by package, and Table 21.2 lists the port functions.



Note: This figure shows a basic port configuration. The configuration differs depending on the ports.

Figure 21.1 Connection diagram for I/O port registers

Table 21.1 I/O port specifications (1 of 3)

Port	Package		Package	
	289, 303 pins	Number of Pins	289, 303 pins (without MIPI)	Number of Pins
PORT0	P000 to P015	16	P000 to P015	16
PORT1	P100 to P115	16	P100 to P115	16
PORT2	P200, P201, P206 to P215	12	P200 to P215	16

Table 21.1 I/O port specifications (2 of 3)

Port	Package		Package	
	289, 303 pins	Number of Pins	289, 303 pins (without MIPI)	Number of Pins
PORT3	P300 to P312	13	P300 to P315	16
PORT4	P400 to P415	16	P400 to P415	16
PORT5	P500 to P515	16	P500 to P515	16
PORT6	P600 to P615*1	16	P600 to P615*1	16
PORT7	P700 to P715	16	P700 to P715	16
PORT8	P800 to P815	16	P800 to P815	16
PORT9	P902 to P915	14	P900 to P915	16
PORTA	PA00 to PA15	16	PA00 to PA15	16
PORTB	PB00 to PB07	8	PB00 to PB07	8
PORTC	PC00 to PC15*1	16	PC00 to PC15*1	16
PORTD	PD00 to PD07	8	PD00 to PD07	8

Table 21.1 I/O port specifications (3 of 3)

Port	Package		Package	
	224 pins	Number of Pins	224 pins (without MIPI)	Number of Pins
PORT0	P000 to P009, P014, P015	12	P000 to P009, P014, P015	12
PORT1	P100 to P107, P112 to P115	12	P100 to P107, P112 to P115	12
PORT2	P200, P201, P206, P208 to P215	11	P200 to P206, P208 to P215	15
PORT3	P300 to P302, P304 to P312	12	P300 to P302, P304 to P315	15
PORT4	P400 to P411, P414, P415	14	P400 to P411, P414, P415	14
PORT5	P503 to P512	10	P503 to P512	10
PORT6	P600 to P604, P608 to P615	13	P600 to P604, P608 to P615	13
PORT7	P700 to P714	15	P700 to P714	15
PORT8	P800 to P804, P808, P809, P813 to P815	10	P800 to P804, P808, P809, P813 to P815	10
PORT9	P902, P905 to P909	6	P902, P905 to P909	6
PORTA	PA00 to PA15	16	PA00 to PA15	16
PORTB	PB00 to PB04	5	PB00 to PB04	5
PORTC	PC11 to PC15	5	PC11 to PC15	5
PORTD	PD00	1	PD00	1

Note 1. For BGA303 package, P603 to P607 and PC00 to PC07 are connected to the SiP Flash memory inside the package.

Table 21.2 I/O port functions (1 of 3)

Port	Port name	Input pull-up	Open-drain output	Drive capability switching	5Vtolerant	I/O	Power supply
PORT0	P000 to P015	✓	✓	Low	—	Input / Output	AVCC0

Table 21.2 I/O port functions (2 of 3)

Port	Port name	Input pull-up	Open-drain output	Drive capability switching	5Vtolerant	I/O	Power supply
PORT1	P100 to P107	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC2
	P108 to P115	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC
PORT2	P200	✓	—	—	—	Input	VCC
	P201	✓	✓	Low	—	Input / Output	VCC
	P202, P203	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC2
	P204, P205	✓	✓	Low, middle, high, high speed high drive	✓	Input / Output	VCC2
	P206 to P213	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC
	P214, P215	✓	—	—	—	Input	VBATT_R
PORT3	P300 to P302, P304 to P312	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC
	P303	✓	✓	Low, middle, high, high speed high drive	✓	Input / Output	VCC
	P313 to P315	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC2
PORT4	P400 to P406, P414, P415	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC
	P407 to P413	✓	✓	Low, middle, high, high speed high drive	✓	Input / Output	VCC
PORT5	P500 to P502	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC2
	P503 to P510, P513	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC
	P511, P512, P514, P515	✓	✓	Low, middle, high, high speed high drive	✓	Input / Output	VCC
PORT6	P600 to P607	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC2
	P608 to P615	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC
PORT7	P700 to P707	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC
	P708 to P715	✓	✓	Low, middle, high, high speed high drive	✓	Input / Output	VCC
PORT8	P800 to P804, P808 to P812	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC2
	P805 to P807, P813	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC
	P814, P815	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC_USB
PORT9	P900, P901	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC2
	P902 to P915	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC
PORTA	PA00 to PA15	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC

Table 21.2 I/O port functions (3 of 3)

Port	Port name	Input pull-up	Open-drain output	Drive capability switching	5Vtolerant	I/O	Power supply
PORTB	PB00 to PB07	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC
PORTC	PC00 to PC10	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC2
	PC11 to PC15	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC
PORTD	PD00 to PD07	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC

Note: ✓: Available
—: Setting prohibited

21.2 Register Descriptions

21.2.1 PCNTR1/PODR/PDR : Port Control Register 1

Base address: PORTm = 0x4040_0000 + 0x0020 × m (m = 0 to 9, A, B, C, D)
PORTm_NS = 0x5040_0000 + 0x0020 × m (m = 0 to 9, A, B, C, D)

Offset address: 0x000 (PCNTR1/PDR)
0x002 (PODR)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	PODR15	PODR14	PODR13	PODR12	PODR11	PODR10	PODR09	PODR08	PODR07	PODR06	PODR05	PODR04	PODR03	PODR02	PODR01	PODR00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PDR15	PDR14	PDR13	PDR12	PDR11	PDR10	PDR09	PDR08	PDR07	PDR06	PDR05	PDR04	PDR03	PDR02	PDR01	PDR00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	PDR15 to PDR00	Pmn Direction 0: Input (functions as an input pin) 1: Output (functions as an output pin)	R/W
31:16	PODR15 to PODR00	Pmn Output Data 0: Low output 1: High output	R/W

Note: S-TYPE-4, P-TYPE-5
Note: m = 0 to 9, A, B, C, D, n = 00 to 15

The Port Control Register 1 (PCNTR1/PODR/PDR) is a 32-bit or 16-bit read/write register that controls port direction and port output data. The PCNTR1 specifies the port direction and output data, and is accessed in 32-bit units. The PDRn (bits [15:0] in PCNTR1) and PODRn (bits [31:16] in PCNTR1) respectively, are accessed in 16-bit units.

PDRn bits (Pmn Direction)

The PDRn bits select the input or output direction for individual pins on the associated port when the pins are configured as general I/O pins. Each pin on port m is associated with a PORTm.PCNTR1.PDRn bit. The I/O direction can be specified in 1-bit unit. Bits associated with non-existent pins are reserved. Reserved bits are read as 0. The write value should be 0. In the case of input only ports, PDRn bits are reserved. See Table 21.2. The PDRn bit in the PORTm.PCNTR1 register serves the same function as the PDR bit in the PFS.PmnPFS register.

PODRn bits (Pmn Output Data)

The PODRn bits hold data to be output from the general I/O pins. Bits of non-existent port m are reserved. Reserved bits are read as 0. The write value should be 0. In the case of input only ports, PODRn bits are reserved. See Table 21.2. The PODRn bit in the PORTm.PCNTR1 register serves the same function as the PODR bit in the PFS.PmnPFS register.

21.2.2 PCNTR2/EIDR/PIDR : Port Control Register 2

Base address: PORTm = 0x4040_0000 + 0x0020 × m (m = 0 to 9, A, B, C, D)
PORTm_NS = 0x5040_0000 + 0x0020 × m (m = 0 to 9, A, B, C, D)

Offset address: 0x004 (PCNTR2/PIDR)
0x006 (EIDR)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	EIDR15	EIDR14	EIDR13	EIDR12	EIDR11	EIDR10	EIDR09	EIDR08	EIDR07	EIDR06	EIDR05	EIDR04	EIDR03	EIDR02	EIDR01	EIDR00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PIDR15	PIDR14	PIDR13	PIDR12	PIDR11	PIDR10	PIDR09	PIDR08	PIDR07	PIDR06	PIDR05	PIDR04	PIDR03	PIDR02	PIDR01	PIDR00
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
15:0	PIDR15 to PIDR00	Pmn State 0: Low level 1: High level	R
31:16	EIDR15 to EIDR00 *2	Port Event Input Data*1 When an ELC_PORTx signal occurs 0: Low input 1: High input	R

Note: S-TYPE-4, P-TYPE-5
Note: m = 0 to 9, A, B, C, D, n = 00 to 15
Note 1. x = 1, 2, 3 or 4 for EIDR only
Note 2. Supported by ports 1, 2, 3 or 4.

The Port Control Register 2 (PCNTR2/EIDR/PIDR) allows read access to the Pmn state and the port event input data using 32-bit or 16-bit access.

The PCNTR2 specifies the Pmn state and the port event input data, and is accessed in 32-bit units.

The PIDRn (bits [15:0] in PCNTR2) and EIDRn (bits [31:16] in PCNTR2) respectively, are accessed in 16-bit units. Bits associated with non-existent pins are reserved. Reserved bits are read as undefined.

PIDRn bits (Pmn State)

The PIDRn bits reflect the individual pin states of the port, regardless of the values set in PmnPFS.PMR and PORTm.PCNTR1.PDRn. The PIDRn bit in the PORTm.PCNTR2 register serves the same function as the PIDR bit in the PFS.PmnPFS register.

A pin state cannot be reflected in PIDRn when one of the following functions is enabled:

- RTC Time Capture input and tamper detection (RTCIC)
- Analog function (ASEL = 1)
- I3C bus interface (I3C_SCL0, I3C_SDA0)
- JTAG/SWD interface (TCK/SWCLK, TMS/SWDIO, TDI, TDO)
- Trace interface (TCLK, TDATAn (n = 0 to 3), SWO)

EIDRn bits (Port Event Input Data)

The EIDRn bits latch a pin state when an ELC_PORTx signal occurs. Pin states can only be input to EIDRn when PmnPFS.PMR and PORTm.PCNTR1.PDRn are 0. When the PmnPFS.ASEL bit is set to 1, the associated pin state is not reflected in EIDRn.

21.2.3 PCNTR3/PORR/POSR : Port Control Register 3

Base address: PORTm = 0x4040_0000 + 0x0020 × m (m = 0 to 9, A, B, C, D)
PORTm_NS = 0x5040_0000 + 0x0020 × m (m = 0 to 9, A, B, C, D)

Offset address: 0x008 (PCNTR3/POSR)
0x00A (PORR)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	PORR15	PORR14	PORR13	PORR12	PORR11	PORR10	PORR09	PORR08	PORR07	PORR06	PORR05	PORR04	PORR03	PORR02	PORR01	PORR00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	POSR15	POSR14	POSR13	POSR12	POSR11	POSR10	POSR09	POSR08	POSR07	POSR06	POSR05	POSR04	POSR03	POSR02	POSR01	POSR00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	POSR15 to POSR00	Pmn Output Set 0: No effect on output 1: High output	W
31:16	PORR15 to PORR00	Pmn Output Reset 0: No effect on output 1: Low output	W

Note: S-TYPE-4, P-TYPE-5
Note: m = 0 to 9, A, B, C, D, n = 00 to 15

The Port Control Register 3 (PCNTR3/PORR/POSR) is a 32-bit or 16-bit write register that controls the setting or resetting of the port output data.

The PCNTR3 controls the setting or resetting of the port output data, and is accessed in 32-bit units.

The POSRn (bits [15:0] in PCNTR3) and the PORRn (bits [31:16] in PCNTR3) respectively, are accessed in 16-bit units.

POSRn bits (Pmn Output Set)

POSR changes PODR when set by a software write. For example, for P100, when PORT1.PCNTR3.POSR00 = 1, PORT1.PCNTR1.PODR00 outputs 1. Bits associated with non-existent pins are reserved. The write value should always be 0. In the case of input only ports, POSRn bits are reserved. See Table 21.2.

PORRn bits (Pmn Output Reset)

PORR changes PODR when reset by a software write. For example, for P100, when PORT1.PCNTR3.PORR00 = 1, PORT1.PCNTR1.PODR00 outputs 0. Bits associated with non-existent pins are reserved. The write value should always be 0. In the case of input only ports, PORRn bits are reserved. See Table 21.2.

Note: When EORRn or EOSRn is set, writing is prohibited to PODRn, PORRn, and POSRn.
Note: PORRn and POSRn should not be set at the same time.

21.2.4 PCNTR4/EORR/EOSR : Port Control Register 4

Base address: PORTm = 0x4040_0000 + 0x0020 × m (m = 1 to 4)
PORTm_NS = 0x5040_0000 + 0x0020 × m (m = 1 to 4)

Offset address: 0x00C (PCNTR4/EOSR)
0x00E (EORR)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	EORR 15	EORR 14	EORR 13	EORR 12	EORR 11	EORR 10	EORR 09	EORR 08	EORR 07	EORR 06	EORR 05	EORR 04	EORR 03	EORR 02	EORR 01	EORR 00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	EOSR 15	EOSR 14	EOSR 13	EOSR 12	EOSR 11	EOSR 10	EOSR 09	EOSR 08	EOSR 07	EOSR 06	EOSR 05	EOSR 04	EOSR 03	EOSR 02	EOSR 01	EOSR 00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	EOSR15 to EOSR00	Pmn Event Output Set When an ELC_PORTx signal occurs 0: No effect on output 1: High output	R/W
31:16	EORR15 to EORR0	Pmn Event Output Reset When an ELC_PORTx signal occurs 0: No effect on output 1: Low output	R/W

Note: S-TYPE-4, P-TYPE-5
Note: m = 1 to 4, n = 00 to 15, x = 1 to 4

The Port Control Register 4 (PCNTR4/EORR/EOSR) is a 32-bit or 16-bit read/write register that controls the setting or resetting of the port output data by an event input from the ELC.

The PCNTR4 controls the setting or resetting of the port output data by an event input from the ELC, and is accessed in 32-bit units.

The EOSRn (bits [15:0] in PCNTR4) and EORRn (bits [31:16] in PCNTR4) respectively, are accessed in 16-bit units.

EOSRn bits (Pmn Event Output Set)

EOSR changes PODR when set because an ELC_PORTx signal occurs. For example, for P100 if PORT1.PCNTR4.EOSR00 is set to 1 when the ELC_PORTx occurs, PORT1.PCNTR1.PODR00 outputs 1. Bits associated with non-existent pins are reserved. The write value should always be 0. For input only ports, EOSRn bits are reserved. See Table 21.2.

EORRn bits (Pmn Event Output Reset)

EORR changes PODR when reset because an ELC_PORTx signal occurs. For example, for P100 if PORT1.PCNTR4.EORR00 = 1 when the ELC_PORTx occurs, PORT1.PCNTR1.PODR00 outputs 0. Bits associated with non-existent pins are reserved. The write value should always be 0. For input only ports, EORRn bits are reserved. See Table 21.2.

Note: When EORRn or EOSRn is set, writing is prohibited to PODRn, PORRn, and POSRn.

Note: EORRn and EOSRn should not be set at the same time.

21.2.5 PmnPFS/PmnPFS_HA/PmnPFS_BY : Port mn Pin Function Select Register (m = 0 to 9, A, B, C, D, n = 00 to 15)

Base address: PFS = 0x4040_0800
PFS_NS = 0x5040_0800

Offset address: 0x000 + 0x040 × m + 0x004 × n (PmnPFS/PmnPFS_HA/PmnPFS_BY)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	PSEL[4:0]					—	—	—	—	—	—	—	PMR
Value after reset:	0	0	0	0*1	0	0	0*1	0*1	0	0	0	0	0	0	0	0*1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	ASEL	ISEL	EOFR[1:0]		DSCR[1:0]		—	—	—	NCODR	—	PCR	—	PDR	PIDR	PODR
Value after reset:	0	0	0	0	0	0*1	0	0	0	0	0	0*1	0	0	x	0

Bit	Symbol	Function	R/W
0	PODR	Port Output Data 0: Low output 1: High output	R/W
1	PIDR	Pmn State 0: Low level 1: High level	R
2	PDR	Port Direction 0: Input (functions as an input pin) 1: Output (functions as an output pin)	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	PCR	Pull-up Control 0: Disable input pull-up 1: Enable input pull-up	R/W
5	—	This bit is read as 0. The write value should be 0.	R/W
6	NCODR	N-Channel Open-Drain Control 0: CMOS output 1: NMOS open-drain output	R/W
9:7	—	These bits are read as 0. The write value should be 0.	R/W
11:10	DSCR[1:0]	Port Drive Capability 0 0: Low drive 0 1: Middle drive 1 0: High speed high drive 1 1: High drive	R/W
13:12	EOFR[1:0]	Event on Falling/Event on Rising*2 0 0: Don't care 0 1: Detect rising edge 1 0: Detect falling edge 1 1: Detect both edges	R/W
14	ISEL	IRQ Input Enable 0: Not used as an IRQn input pin 1: Used as an IRQn input pin	R/W
15	ASEL	Analog Input Enable 0: Not used as an analog pin 1: Used as an analog pin	R/W
16	PMR	Port Mode Control 0: Used as a general I/O pin 1: Used as an I/O port for peripheral functions	R/W
23:17	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
28:24	PSEL[4:0]	Peripheral Select These bits select the peripheral function. For individual pin functions, see the associated tables in this chapter.	R/W
31:29	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

Note 1. The initial value of P201, P208, P209. P210, P211, P814 and P815 is not 0x0000_0000. P201 is 0x0000_0010, P208 is 0x0001_0010, P209 is 0x0001_0400, P210 is 0x0001_0410, P211 is 0x0001_0010, P814 is 0x1301_0000 and P815 is 0x1301_0000.

Note 2. Supported by PORTn (n = 1 to 4).

Port mn Pin Function Select Register (PmnPFS/PmnPFS_HA/PmnPFS_BY) is a 32-bit, 16-bit, or 8-bit read/write control register that selects the port mn pin function, and is accessed in 32-bit units. PmnPFS_HA (PmnPFS [15:0] bits) is accessed in 16-bit units. PmnPFS_BY (PmnPFS[7:0] bits) is accessed in 8-bit units.

The available Port mn pin depends on the product. For details, see Table 21.1

PODR bit (Port Output Data), PIDR bit (Port State), PDR bit (Port Direction)

The PDR, PIDR, and PODR bits serve the same function as the PCNTR. When these bits are read, the PCNTR value is read.

PCR bit (Pull-up Control)

The PCR bit enables or disables an input pull-up resistor on the individual port pins. When a pin is in the input state with the associated bit in PmnPFS.PCR set to 1, the pull-up resistor connected to the pin is enabled. When a pin is set as a general port output pin, or a peripheral function output pin, the pull-up resistor for the pin is disabled regardless of the PCR setting. The pull-up resistor is also disabled in the reset state. Bits associated with non-existent pins are reserved. Reserved bits are read as 0. The write value should be 0.

NCODR bit (N-Channel Open-Drain Control)

The NCODR bit specifies the output type for the port pins. Bits associated with non-existent pins are reserved. Reserved bits are read as 0. The write value should be 0.

DSCR[1:0] bits (Port Drive Capability)

The DSCR[1:0] bits switch the drive capacity of the port. If the drive capacity of a pin is fixed, the associated bit is a read/write bit, but the drive capacity cannot be changed. Bits associated with non-existent pins are reserved. Reserved bits are read as 0. The write value should be 0.

EOFR[1:0] bits (Event on Falling/Event on Rising)

The EOFR[1:0] bits select the edge detection method for the port group input signal. These bits support rising, falling, or both edge detections. When the EOFR[1:0] bits are set to 01b, 10b, or 11b, the input enable of the I/O cell is asserted. Following that, the event pulse is input from the external pin, and the GPIO outputs the event pulse to the ELC. Bits associated with non-existent pins are reserved. Reserved bits are read as 0. The write value should be 0.

ISEL bit (IRQ Input Enable)

The ISEL bit specifies IRQ input pins. This setting can be used in combination with the peripheral functions, although an IRQn (external pin interrupt) of the same number must only be enabled for one pin. The ISEL bit for an unspecified IRQn is reserved.

ASEL bit (Analog Input Enable)

The ASEL bit specifies analog pins. When a pin is set as an analog pin by this bit:

1. Specify it as a general I/O port in the Port Mode Control bit (PmnPFS.PMR).
2. Disable the pull-up resistor in the Pull-up Control bit (PmnPFS.PCR).
3. Specify the input in the Port Direction bit (PmnPFS.PDR). The pin state cannot be read at this point. The PmnPFS register is protected by the Write-Protect Register for Non-secure (PWPR_NS). Release write-protect before modifying the register.

The ASEL bit for an unspecified analog I/O pin is reserved.

PMR bit (Port Mode Control)

The PMR bit specifies the port pin function. Bits associated with non-existent pins are reserved. The write value should be 0.

PSEL[4:0] bits (Peripheral Select)

The PSEL[4:0] bits assign the peripheral function.

21.2.6 PWPR_NS : Write-Protect Register for Non-secure

Base address: PFS_NS = 0x5040_0800

Offset address: 0x50C

Bit position:	7	6	5	4	3	2	1	0
Bit field:	B0WI	PFSWE	—	—	—	—	—	—
Value after reset:	1	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	—	These bits are read as 0. The write value should be 0.	R/W
6	PFSWE	PmnPFS Register Write Enable 0: Writing to the PmnPFS register is disabled 1: Writing to the PmnPFS register is enabled	R/W
7	B0WI	PFSWE Bit Write Disable 0: Writing to the PFSWE bit is enabled 1: Writing to the PFSWE bit is disabled	R/W

Note: S-TYPE-7, P-TYPE-2

Note: The access to this register is not controlled by any security attribution register.

PFSWE bit (PmnPFS Register Write Enable)

Writing to the PmnPFS register is enabled only when the PFSWE bit is set to 1. You must first write 0 to the B0WI bit before setting PFSWE to 1.

B0WI bit (PFSWE Bit Write Disable)

Writing to the PFSWE bit is enabled only when the B0WI bit is set to 0.

21.2.7 PFENET : Ethernet Control Register

Base address: PFS = 0x4040_0800
PFS_NS = 0x5040_0800

Offset address: 0x500

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	ETHIOMD1	ETHIOMD0	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0. The write value should be 0.	R/W
4	ETHIOMD0	Ethernet I/O Mode Setting ch0 0: GMII mode (ESWM channel 0) 1: MII mode (ESWM channel 0)	R/W
5	ETHIOMD1	Ethernet I/O Mode Setting ch1 0: GMII mode (ESWM channel 1) 1: MII mode (ESWM channel 1)	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note: The access to the ETHIOMD0/1 bit is controlled by the PSARC.PSARC30 bit and PPARC.PPARC30 bit.

ETHIOMD0/ETHIOMD1 bit (Ethernet I/O Mode Setting ch0 and Ethernet I/O Mode Setting ch1)

When the transmit data output ports of ESWM are used as GMII/RGMII ports, the drive capability can be set by the ETHIOMDn bit and DSCR[1:0] bit. The details of the setting are given by the following table:

Interface type	PSEL[4:0]	ETHIOMDn	VCC voltage [V]	DSCR [1:0]	Drive capability
MII only	0x16	1b	3.6-2.7	01b	Middle drive
MII/GMII		0b	3.6-3.0	01b	RGMII/3.3 V
			2.7-2.3	11b	RGMII/2.5 V
RMII	0x17	—	3.6-2.7	11b	High drive
RGMII	0x18		3.6-3.0	01b	RGMII/3.3 V
			2.7-2.3	11b	RGMII/2.5 V

If the settings of the ETHIOMDn bit and DSCR[1:0] bit other than those in the above table are performed, the electrical characteristics cannot be guaranteed.

21.2.8 PWPR_S : Write-Protect Register for Secure

Base address: PFS = 0x4040_0800

Offset address: 0x514

Bit position:	7	6	5	4	3	2	1	0
Bit field:	B0WI	PFSWE	—	—	—	—	—	—
Value after reset:	1	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	—	These bits are read as 0. The write value should be 0.	R/W
6	PFSWE	PmnPFS Register Write Enable 0: Disable writes to the PmnPFS register 1: Enable writes to the PmnPFS register	R/W
7	B0WI	PFSWE Bit Write Disable 0: Enable writes the PFSWE bit 1: Disable writes to the PFSWE bit	R/W

Note: S-TYPE-6, P-TYPE-2

PFSWE bit (PmnPFS Register Write Enable)

Writing to the PmnPFS register of the IO port pin set as secure by the PmSAR register is enabled only when the PFSWE bit is set to 1. You must first write 0 to the B0WI bit before setting PFSWE to 1.

B0WI bit (PFSWE Bit Write Disable)

Writing to the PFSWE bit is enabled only when the B0WI bit is set to 0.

21.2.9 PmSAR : Port Security Attribution Register (m = 0 to 9, A, B, C, D)

Base address: PFS = 0x4040_0800
PFS_NS = 0x5040_0800

Offset address: 0x530 + 0x004 × m

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PMNSA[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	PMNSA[15:0]	Pmn Security Attribution Target I/O port pin : Pmn 0: Secure 1: Non-secure	R/W

Note: S-TYPE-1, P-TYPE-1
Note: This register is write-protected by PRCR register.
Note: m = 0 to 9, A, B, C, D, n = 00 to 15

Port Security Attribution Register is a 16-bit register that setting the Security Attribution of each port, the registers are accessed only in 16-bit units.

PMNSA[15:0] bits (Pmn Security Attribution)

The PmnSA bit specifies the Security Attribution of Pmn.

21.2.10 LVOCR : Low Voltage Operation Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xAB0

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	LVO1E	LVO0E
Value after reset:	0	0	0	0	0	0	1	1

Bit	Symbol	Function	R/W
0	LVO0E	Low Voltage Operation 0 Enable 0: Disable 1: Enable	R/W
1	LVO1E	Low Voltage Operation 1 Enable 0: Disable 1: Enable	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

Low voltage operation control register controls the IO when VCC/VCC2 is lower than 2.7V.

LVO0E bit (Low Voltage Operation 0 Enable)

When VCC is lower than 2.7V, set the LVO0E bit to 1.

LVO1E bit (Low Voltage Operation 1 Enable)

When VCC2 is lower than 2.7V, set the LVO1E bit to 1.

21.3 Operation

21.3.1 General I/O Ports

All pins except P208 to P211, P814 and P815 operate as general I/O ports after reset. General I/O ports are organized as 16 bits per port and can be accessed by port with the Port Control Registers (PCNTRn, where n = 1 to 4), or by individual pins with the Port mn Pin Function Select register. For details on these registers, see section 21.2. Register Descriptions.

Each port has the following bits:

- Port Security Attribution register (PmSAR) (m = 0 to 9, A, B, C, D), which indicates the security attribution.
- Port Direction bit (PDRn), which selects input or output direction
- Port Output Data bit (PODRn), which holds data for output
- Port Input Data bit (PIDRn), which indicates the pin states
- Event Input Data bit (EIDRn), which indicates the pin state when an ELC_PORTn (n = 1, 2, 3 or 4) signal occurs
- Port Output Set bit (POSRn), which indicates the output value when a software write occurs
- Port Output Reset bit (PORRn), which indicates the output value when a software write occurs
- Event Output Set bit (EOSRn), which indicates the output value when an ELC_PORTn (n = 1, 2, 3 or 4) signal occurs
- Event Output Reset bit (EORRn), which indicates the output value when an ELC_PORTn (n = 1, 2, 3 or 4) signal occurs.

21.3.2 Port Function Select

The following port functions are available for configuring each pin:

- Security function: Security attribution for each pins
- I/O configuration: CMOS output or NMOS open-drain output, pull-up control, and drive capacity
- General I/O port: Port direction, output data setting, and read input data
- Alternate function: Configured function mapping to the pin.

Each pin is associated with a Port mn Pin Function Select register (PmnPFS), which includes the associated PODR, PIDR, and PDR bits. In addition, the PmnPFS register includes the following:

- PCR: Pull-up resistor control bit that turns the input pull-up MOS on or off
- NCODR: N-channel open-drain control bit that selects the output type for each pin
- DSCR[1:0]: Drive capacity control bit that selects the drive capacity
- EOFR[1:0]: For selecting the edge of the event that input from the port group
- ISEL: IRQ input enable bit to specify an IRQ input pin
- ASEL: Analog input enable bit to specify an analog pin
- PMR: Port mode bit to specify the pin function of each port
- PSEL[4:0]: Port function select bits to select the associated peripheral function.

These configurations can be made by a single-register access to the Port mn Pin Function Select register. For details, see section 21.2.5. PmnPFS/PmnPFS_HA/PmnPFS_BY : Port mn Pin Function Select Register (m = 0 to 9, A, B, C, D, n = 00 to 15).

21.3.3 Port Group Function for ELC

In the MCU, PORT1 to PORT4 are assigned for the ELC port group function.

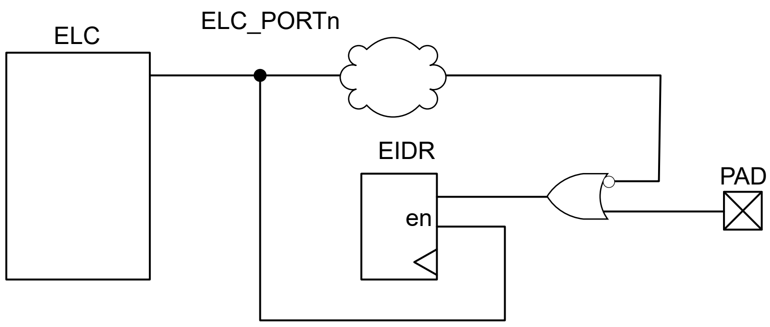
21.3.3.1 Behavior When ELC_PORTn (n = 1, 2, 3 or 4) is Input from ELC

The MCU supports the two functions described in this section when an ELC_PORTn (n = 1, 2, 3 or 4) signal comes from the ELC.

(1) Input to EIDR

For the GPI function (PDR = 0 and PMR = 0 in the PmnPFS register), when an ELC_PORTn (n = 1, 2, 3 or 4) signal comes from the ELC, the input enable of the I/O cell is asserted, and data from the external pins is read into the EIDR bit. See Figure 21.2

For the GPO function (PDR = 1) or the peripheral mode (PMR = 1), 0 is input into the EIDR bit from the external pins.



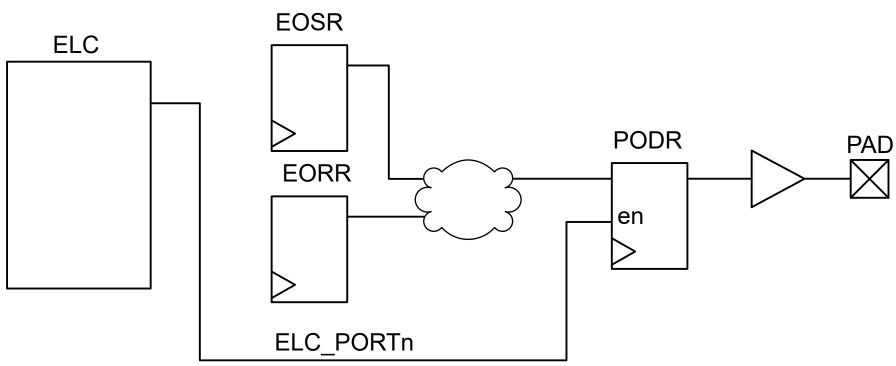
Note: n = 1, 2, 3 or 4

Figure 21.2 Event ports input data

(2) Output from PODR by EOSR and EORR

When an ELC_PORTn (n = 1, 2, 3 or 4) signal occurs, the data is output from the PODR to the external pin based on the settings in the EOSR and EORR registers.

- If EOSR is set to 1, when an ELC_PORTn (n = 1, 2, 3 or 4) signal occurs, the PODR register outputs 1 to the external pin. Otherwise, when EOSR = 0, the PODR value is retained.
- If EORR is set to 1, when ELC_PORTn (n = 1, 2, 3 or 4) signal occurs, the PODR register outputs 0 to the external pin. Otherwise, when EORR = 0, the PODR value is retained.



Note: n = 1, 2, 3 or 4

Figure 21.3 Event ports output data

21.3.3.2 Behavior When an Event Pulse is Output to ELC

To output the event pulse from the external pins to the ELC, set the EOFR[1:0] bits in the PmnPFS register. For details, see section 21.2.5. PmnPFS/PmnPFS_HA/PmnPFS_BY : Port mn Pin Function Select Register (m = 0 to 9, A, B, C, D, n = 00 to 15). When the EOFR[1:0] bits are set, the input enable of the I/O cell is asserted.

Data from the external pin is the input. For example, for Port 1, when the data is input from P100 to P115, the data of those 16 pins is organized by OR logic. This data is formed into a one-shot pulse that goes to the ELC. The operation of Port n (n = 2 to 4) is also the same as Port 1. See Figure 21.4.

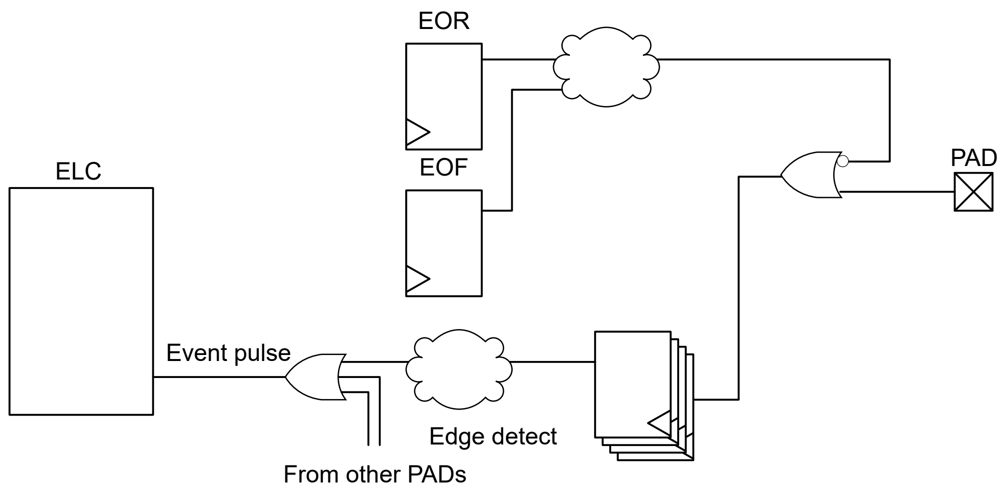


Figure 21.4 Generation of event pulse

21.4 Handling of Unused Pins

Table 21.3 shows how to handle unused pins.

Table 21.3 Handling of unused pins (1 of 2)

Pin name	Description
MD	Use as a mode selection pin
RES	Connect to VCC through a resistor
USB_DP, USB_DM	• Preconditions: – VCC_USB: Connect to VCC – VSS_USB: Connect to VSS – Set the module-stop state for USBFS (MSTPCRB.MSTPB11 = 1) • Processing details: – When both P814PFS.PMR and P815PFS.PMR bits are set to 1, keep these pins open. – When P814PFS.PMR or P815PFS.PMR bit is set to 0, configure it in the same way as P814, P815.
P200/NMI	Connect to VCC through a resistor
EXTAL	When the main clock oscillator is not used, set the MOSCCR.MOSTP bit to 1 (general port P212). When this pin is not used as port P212, configure it in the same way as Other ports.
XTAL	When the main clock oscillator is not used, set the MOSCCR.MOSTP bit to 1 (general port P213). When the external clock is input to the EXTAL pin, the XTAL pin functions as P213. When this pin is not used as port P213, configure it in the same way as Other ports.
XCIN/EXCIN	When the sub clock oscillator is not used, set the SOSCCR.SOSTP bit to 1 and set the SOMCR.SOSEL bit to 0 (general port P215). When this pin is not used as port P215, configure it in the same way as P214, P215.

Table 21.3 Handling of unused pins (2 of 2)

Pin name	Description
XCOUT	When the sub clock oscillator is not used, set the SOSCCR.SOSTP bit to 1 (general port P214). When the external clock is input to the EXCIN pin, the XCOUT pin functions as P214. When this pin is not used as port P214, configure it in the same way as P214, P215.
P000 to P015	• If the direction is set to input (PCNTR1.PDRn = 0), connect the associated pin to AVCC0 through a resistor or to AVSS0 through a resistor *1 • If the direction is set to output (PCNTR1.PDRn = 1), keep pin open *1
P100 to P107, P202 to P205, P313 to P315, P500 to P502, P600 to P607, P800 to P804, P808 to P812, P900, P901, PC00 to PC10	• If the direction is set to input (PCNTR1.PDRn = 0), connect the associated pin to VCC2 through a resistor or to VSS through a resistor *1 • If the direction is set to output (PCNTR1.PDRn = 1), keep pin open *1
P214, P215	Connect to VSS through a resister
P814, P815	• If the direction is set to input (PCNTR1.PDRn = 0), connect the associated pin to VCC_USB through a resistor or to VSS through a resistor *1 • If the direction is set to output (PCNTR1.PDRn = 1), keep pin open *1
Other ports	• If the direction is set to input (PCNTR1.PDRn = 0), connect the associated pin to VCC through a resistor or to VSS through a resistor *1 *2 • If the direction is set to output (PCNTR1.PDRn = 1), keep pin open *1 *3
USBHS_DP, USBHS_DM, USBHS_RREF	• Preconditions: – AVCC_USBHS = VCC_USBHS: Connect to VCC – VSS1_USBHS = VSS2_USBHS: Connect to VSS – Set the module-stop state for USBHS (MSTPCRB.MSTPB12 = 1) • Processing details: – USBHS_DP, USBHS_DM, and USBHS_RREF: Open
VREFH0, VREFH	Connect to AVCC0
VREFL0, VREFL	Connect to AVSS0
MIPI_DL0_N, MIPI_DL1_N, MIPI_DL0_P, MIPI_DL1_P, MIPI_CL_N, MIPI_CL_P	• Preconditions: – VCC18_MIPI: Open – AVCC_MIPI: Connect to VCC – VSS_MIPI: Connect to VSS – Set the module-stop state for MIPI DSI (MSTPCRC.MSTPC10 = 1) – Set the module-stop state for MIPI CSI (MSTPCRC.MSTPC17 = 1) • Processing details: – MIPI_DL0_N, MIPI_DL1_N, MIPI_DL0_P, MIPI_DL1_P, MIPI_CL_N and MIPI_CL_P: Open
VLO	Keep pin open*4
VBATT	Connect to VCC

Note 1. Clear the PmnPFS.PMR, PmnPFS.ISEL, PmnPFS.PCR, and PmnPFS.ASEL bits to 0.
Note 2. P208, P210 and P211 are recommended for pull up VCC through a resistor, because these pins are input pull-up enabled from the initial value (PmnPFS.PCR = 1).
Note 3. P209 is recommended for setting the direction to output (PCNTR1.PDRn = 1), because this pin is output from the initial value.
Note 4. For more information, see section 69, Internal Voltage Regulator

21.5 Usage Notes

21.5.1 Procedure for Specifying the Pin Functions

To specify the I/O pin functions:

1. Clear the B0WI bit in the PWPR_NS register. This enables writing to the PFSWE bit in the PWPR_NS register.*1
2. Set 1 to the PFSWE bit in the PWPR_NS register. This enables writing to the PmnPFS register.*1
3. Clear the Port Mode Control bit in the PMR to 0 for the target pin to select the general I/O port.
4. Specify the I/O function for the pin through the PSEL[4:0] bits settings in the PmnPFS register.

5. Set the PMR bit to 1 as required to switch to the selected I/O function for the pin.
6. Clear the PFSWE bit in the PWPR_NS register. This disables writing to the PmnPFS register.*1
7. Set 1 to the B0WI bit in the PWPR_NS register. This disables writing to the PFSWE bit in the PWPR_NS register.*1

Note 1. When the security attribution of Pmn is set to 0, set the PWPR_S register to write to the PmnPFS register.

21.5.2 Procedure for Using Port Group Input

To use the port group input (port n (n = 1 to 4)):

1. Set the ELSRx.ELS[8:0] bits to all 0 to ignore unexpected pulses. For more information, see section 20, Event Link Controller (ELC).
2. Set the EOFR[1:0] bits of the PmnPFS register to specify the rising, falling, or both edge detections.
3. Execute a dummy read or wait for a short time, for example 100 ns. Ignoring of unexpected pulses depends on the initial value of the external pin.
4. Set the ELSRx.ELS[8:0] bits to enable the event signals.

21.5.3 Port Output Data Register (PODR) Summary

This register outputs data as follows:

1. Outputs 0 if PCNTR4.EORR is set to 1 when ELC_PORTn (n = 1, 2, 3 or 4) signal occurs.
2. Outputs 1 if PCNTR4.EOSR is set to 1 when ELC_PORTn (n = 1, 2, 3 or 4) signal occurs.
3. Outputs 0 if PCNTR3.PORR is set to 1.
4. Outputs 1 if PCNTR3.POSR is set to 1.
5. Outputs 0 or 1 because PCNTR1.PODRn is set.
6. Outputs 0 or 1 because PmnPFS.PODRn is set.

Numbers in this list correspond to the priority for writing to the PODRn. For example, if 1. and 3. from the list occur at the same time, the higher priority event 1. is executed.

21.5.4 Notes on Using Analog Functions

To use an analog function, set the Port Mode Control bit (PMR) and the Port Direction bit (PDRn) to 0 so that the pin acts as a general input port. Next, set the Analog Input Enable bit (ASEL) in the Port mn Pin Function Select Register (PmnPFS.ASEL) to 1.

21.5.5 I/O Buffer Specification

The P402, P403, and P404 can be used as the RTC input, tamper detection and other peripheral functions. Table 21.4 lists the P402, P403, P404 specifications.

Table 21.4 P402, P403, P404 specifications

I/O port	RTC and tamper detection			Other peripheral	
	RTC and tamper detection input enable register	RTC	Tamper detection	Other peripheral enable register	ADC16H, GPT, CAC, ESWM, CANFD, SCI, SSIE, SDHI and interrupt
P402	VBTICTLR.VCH0INEN	RTCIC0	RTCIC0	P402PFS.PSEL and PMR	For details, see section 21.6. Peripheral Select Settings for Each Product.
P403	VBTICTLR.VCH1INEN	RTCIC1	RTCIC1	P403PFS.PSEL and PMR	
P404	VBTICTLR.VCH2INEN	RTCIC2	RTCIC2	P404PFS.PSEL and PMR	

These RTC inputs and tamper detection are controlled by the VBTICTLR register and this register has the highest priority for selecting the functions.

P402, P403, and P404 can be used as IRQn-DS (n = 4, 14, 15) whether RTC inputs and tamper detection are selected or not. When using these interrupts, set the interrupt procedure after setting the VBTICTLR register. See section 12.2.8. VBTICTLR : VBATT Input Control Register.

See Figure 21.5.

The VBTICTLR register is only initialized by VBATT_POR reset. Therefore, when not using the RTC inputs and tamper detection, the associated bit of VBTICTLR register must be set to 0 after except VBATT_POR reset.

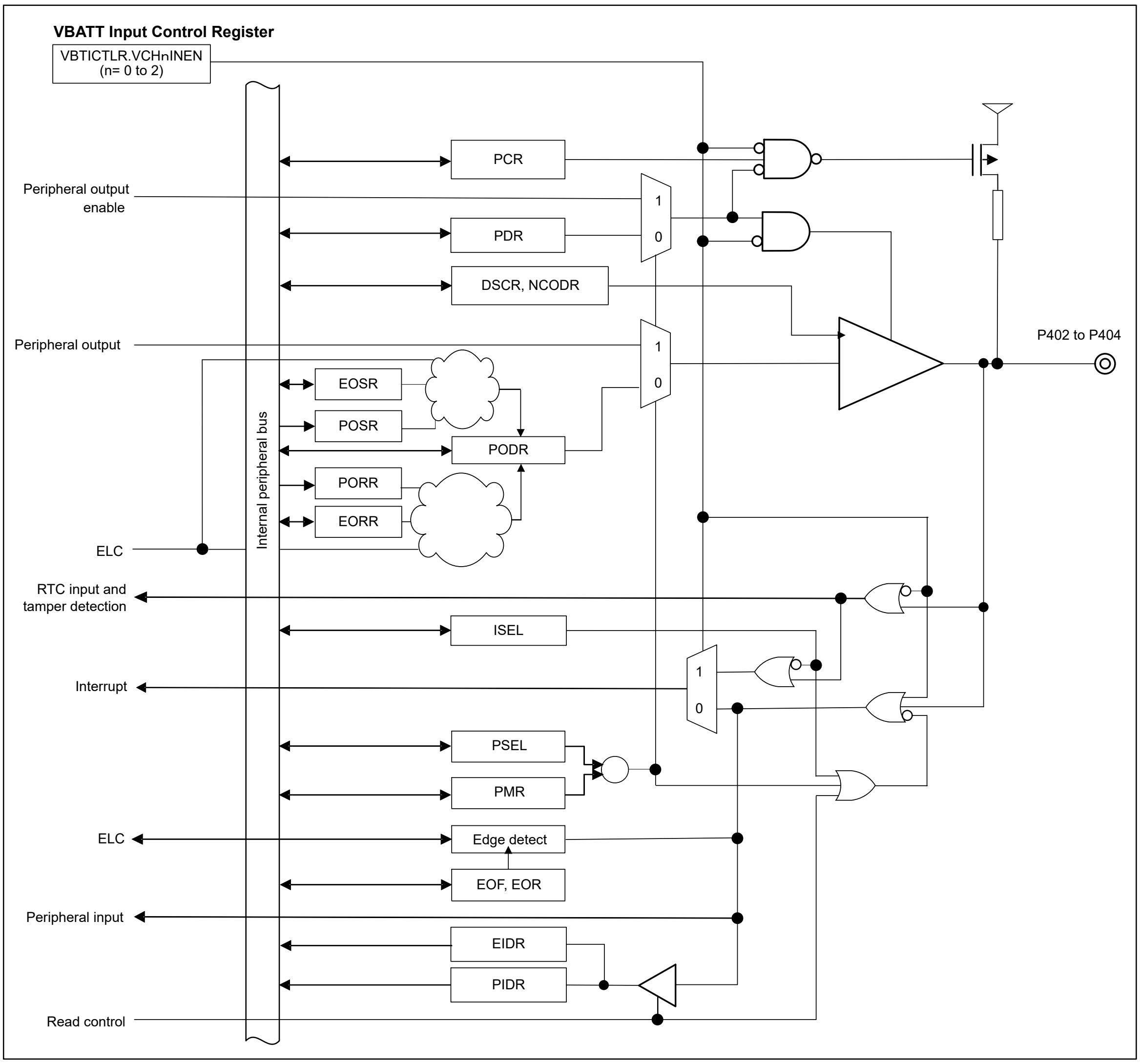


Figure 21.5 P402, P403, P404 diagram

21.5.6 Selecting Pins USB_DP and USB_DM

Pins USB_DP and USB_DM are shared with pins P814 and P815, respectively. Pins USB_DP and P814 can be set with the PFS.P814PFS.PMR bit, and pins USB_DM and P815 can be set with the PFS.P815PFS.PMR bit. Table 21.5 shows setting values of bits PFS.P814PFS.PMR and PFS.P815PFS.PMR with each selected pin.

Table 21.5 Selecting the USB/PORT pins (1 of 2)

PMR Bits Settings		Pins Selected	
P814PFS.PMR bit	P815PFS.PMR bit	P814/USB_DP Pin	P815/USB_DM Pin
0	0	P814	P815
0	1	P814	P815

Table 21.5 Selecting the USB/PORT pins (2 of 2)

PMR Bits Settings		Pins Selected	
P814PFS.PMR bit	P815PFS.PMR bit	P814/USB_DP Pin	P815/USB_DM Pin
1	0	P814	P815
1	1	USB_DP	USB_DM

Note: When using P814/USB_DP and P815/USB_DM as GPIO pins (P814 and P815), use the USB related registers with their initial values.
Note: When using P814/USB_DP and P815/USB_DM as USB pins (USB_DP and USB_DM), use the GPIO related registers for P814 and P815 with their initial values.
Note: When using P814/USB_DP and P815/USB_DM as GPIO pins or USB pins, set these pins only once after a reset.

21.5.7 Pull-up/Pull-down Setting for P814 and P815 at USBFS/GPIO Function

When the GPIO function of P814 and P815 is used, the Pull-up/Pull-down function of USBFS register affects to P814 and P815. Therefore, before using GPIO function, the pull-up and pull-down control of USBFS needs to be disabled. The control register is SYSCFG.DPRPU and SYSCFG.DRPD bits.

21.5.8 Notes on using P215 and P214

Table 21.6 shows the pin selection and notes for P215 and P214.

Table 21.6 Pin selection and notes for P215 and P214

No.	P215	P214	Description
1	XCIN	XCOUT	Permitted*3
2	EXCIN*1	General purpose input*1	Permitted*4
3	EXCIN*1	Unused*2	Permitted*4
4	General purpose input*1	General purpose input*1	Permitted*5
5	General purpose input*1	Unused*2	Permitted*5
6	Unused*2	General purpose input*1	Permitted*6
7	Unused*2	Unused*2	Permitted*6

Note 1. Do not input a voltage higher than VCC to this pin when VCC > $V_{DETBATT}$.
Do not input a voltage higher than V_{BATT} to this pin when VCC ≤ $V_{DETBATT}$.
Note 2. Unused pin should be connected to VSS through a resister.
Note 3. Set the SOMCR.SOSEL bit to 0 and set SOSCCR.SOSTP bit to 0
Note 4. Set the SOMCR.SOSEL bit to 1 and set SOSCCR.SOSTP bit to 0
Note 5. Set the SOMCR.SOSEL bit to 0 and set SOSCCR.SOSTP bit to 1
Note 6. Set the SOSCCR.SOSTP bit to 1.

21.6 Peripheral Select Settings for Each Product

This section describes the pin function select configuration using the PmnPFS register. Some pin names have added _A, _B, or _C suffixes. When assigning IIC, SPI, SSIE, and SDHI functionality, select the functional pins having the same suffix. The other pins can be selected regardless of the suffix. Assigning the same function to two or more pins simultaneously is prohibited.

1. In Pmn pin function select register(PmnPFS), the PSEL bits have to be set when the PMR bit of the target pin is 0. If the PSEL bits are set when the PMR bit is 1, the unexpected edges may be input at the input function or the unexpected pulses may be output to the external pin at the output function.
2. Only the allowed values (functions) should be specified in the PSEL bits of PmnPFS register. If a value which is not allowed for the register is specified, the correct operation is not guaranteed.
3. The single function should not be assigned to the multiple pins by PmnPFS register. When the GPT1, GPT5, SCI3, IIC0 or SPI0 are configured as secure and these pin function is being assigned to the pin which security attribution is set as secure by the PmSAR register, the write access to the PSEL bits for setting same function as secure pin in other pins is ignored when the security attribution of that pin is non-secure. For example, if the PSARE.PSARE30 bit is 0 (GPT1

is secure) and the P209PFS.PSEL bits is 00011b (pin function is GTIOC1A) and the P2SAR.P209SA bit is 0 (P209 is secure), the write 00011b to the P405PFS.PSEL bits is ignored when the P4SAR.P405SA bit is 1 (P405 is non-secure).

4. The PORT0, PORT5 and PORT8 have the analog functions such as A/D converter. When these pins are used as an analog function, for avoiding the loss of resolution, the PMR bit should be set to 0 and PDR bit should be set to 0. After that, ASEL bit should be set to 1

Table 21.7 Register settings for input/output pin function (PORT0)

Pin	PSEL[4:0] settings	Function	Pin														
P006			P000	P001	P002	P003	P004	P005	P007	P008	P009	P010	P011	P012	P013	P014	P015
Hi-Z	00000b (value after reset)	Hi-Z/ JTAG/ SWD	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
AN006/ IVCMP 2	ASEL bit		AN000/ IVCMP 2	AN001/ IVCMP 3	AN002/ IVCMP 2	AN003/ IVCMP 3	AN004/ IVCMP 2	AN005/ IVCMP 3	AN007/ IVCMP 3	AN008/ IVREF0	AN009/ IVREF1	AN010	AN011	AN012	AN013	AN014/ DA0/ IVCMP 0	AN015/ DA1/ IVCMP 0
IRQ11-DS	ISEL bit		IRQ6-DS	IRQ7-DS	IRQ8-DS	IRQ29	IRQ9-DS	IRQ10-DS	IRQ28	IRQ12-DS	IRQ13-DS	IRQ14	IRQ16	IRQ15	IRQ14	IRQ27	IRQ13
✓	PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
✓	NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
L	DSCR[1:0] bit	Drive capability control*1	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
✓	289, 303 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
✓	289, 303 pins w/o MIPI product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
✓	224 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	—	—	—	—	✓	✓
✓	224 pins w/o MIPI product		✓	✓	✓	✓	✓	✓	✓	✓	✓	—	—	—	—	✓	✓

✓: Available
—: Setting prohibited

Note 1. The drive strength of this port cannot be controlled by PmnPFS.DSCR[1:0] bits.

Table 21.8 Register settings for input/output pin function (PORT1) (1 of 2)

PSEL[4:0] settings	Function	Pin															
		P100	P101	P102	P103	P104	P105	P106	P107	P108	P109	P110	P111	P112	P113	P114	P115
00000b (value after reset)	Hi-Z/ JTAG/ SWD	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
00001b	AGT	AGTIO 0	AGTEE 0	AGTO0	—	—	—	AGTOB 0	AGTOA 0	—	—	—	—	—	—	—	—
00010b	GPT*2	GTETR GA	GTETR GB	GTOWL O	GTOW UP	GTETR GB	—	GTOWL O	GTOW UP	—	—	—	—	GTETR GA	GTETR GB	GTETR GC	GTETR GD
00011b	GPT*2	GTIOC 8B	GTIOC 8A	GTIOC 2B	GTIOC 2A	GTIOC 1B	GTIOC 1A	GTIOC 8B	GTIOC 8A	GTIOC 10B	GTIOC 10A	GTIOC 9B	GTIOC 9A	GTIOC 3B	GTIOC 2A	GTIOC 2B	GTIOC 5A
00100b	SCI*1	—	—	—	—	—	CTS_R TS8_B/ SS8_B	CTS8_ B	CTS4_ A	—	—	—	—	TXD0_ A/ SDA0_ A/ MOSI0 _A	RXD0_ A/ SCL0_ A/ MISO0 _A	CTS_R TS0_A/ SS0_A	CTS0_ A
00101b	SCI*1	SCK9_ A	RXD9_ A/ SCL9_ A/ MISO9 _A	TXD9_ A/ SDA9_ A/ MOSI9 _A	CTS_R TS9_A/ SS9_A	CTS9_ A	—	—	—	—	—	—	—	—	—	—	—
00110b	SPI*1	MISOB _A	MOSIB _A	RSPCK B_A	SSLB0 _A	SSLB1 _A	SSLB2 _A	SSLB3 _A	—	—	—	—	—	SSLA2 _B	SSLA1 _B	SSLA0 _B	MOSIA _B
01010b	CAC/ ADC16H	—	—	ADTRG 0	AD1FL AG1	AD0FL AG1	ADSYN C	ADST1	ADST0	—	—	—	—	ADST0	ADST1	ADSYN C	AD0FL AG1
01011b	BUS	—	—	—	—	—	—	—	—	—	—	—	—	D3/DQ3	D4/DQ4	D5/DQ5	D6/DQ 6
01101b	SCI	—	—	—	—	—	DE8	—	—	—	—	—	—	—	—	DE0	—
01110b	SCI	DE9	—	—	DE9	—	—	—	—	—	—	—	—	—	—	—	—
10000b	CANFD	—	—	CRX0	CTX0	—	—	—	—	—	—	—	—	—	—	—	—

Table 21.8 Register settings for input/output pin function (PORT1) (2 of 2)

PSEL[4:0] settings	Function	Pin P100	P101	P102	P103	P104	P105	P106	P107	P108	P109	P110	P111	P112	P113	P114	P115
10010b	SSIE*1	—	—	—	—	—	—	—	—	—	—	—	—	SSIBCK0_B	SSILRCK0/SSIFS0_B	SSIRXD0_B	SSITXD0_B
10101b	SDHI*1	—	—	—	—	—	—	—	—	SD0DAT6_C	SD0DAT5_C	SD0DAT4_C	SD0DAT3_C	SD0DAT4_B	SD0DAT5_B	SD0DAT6_B	SD0DAT7_B
10110b	ESWM (GMII/MII)	GPTP_MATCH1	GPTP_CAPTURE1	—	GPTP_PPS0	GPTP_MATCH0	GPTP_CAPTURE0	ET1_LINKSTA	ET1_INT	—	—	—	—	—	—	—	—
11001b	GLCDC	—	—	—	—	—	—	—	—	—	—	—	—	LCD_DATA10_A	LCD_DATA9_A	LCD_DATA8_A	LCD_DATA7_A
11100b	OSPI	OM_0_SIO0	OM_0_SIO3	OM_0_SIO4	OM_0_SIO2	OM_0_CS1	OM_0_ECSINT1	OM_0_RESET	OM_0_CS0	—	—	—	—	—	—	—	—
11110b	ULPT	—	—	—	—	—	ULPTO1-DS	ULPTEE1-DS	—	—	—	—	—	ULPTOB0-DS	ULPTOA0-DS	—	—
ASEL bit		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
ISEL bit		IRQ2	IRQ1	IRQ17	IRQ16	IRQ1	IRQ0	IRQ16	IRQ31	IRQ24	IRQ23	IRQ20	IRQ19	IRQ27	IRQ28	IRQ30-DS	IRQ31-DS
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
DSCR[1:0] bit	Drive capability control	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH
289, 303 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
289, 303 pins w/o MIPI product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		✓	✓	✓	✓	✓	✓	✓	✓	—	—	—	—	✓	✓	✓	✓
224 pins w/o MIPI product		✓	✓	✓	✓	✓	✓	✓	✓	—	—	—	—	✓	✓	✓	✓

✓: Available
—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance "_A", "_B", or "_C" to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK}).

Table 21.9 Register settings for input/output pin function (PORT2) (1 of 2)

PSEL[4:0] settings	Function	Pin P200*4	P201	P202	P203	P204	P205	P206	P207	P208	P209	P210	P211	P212	P213	P214	P215
00000b (value after reset)	Hi-Z/JTAG/SWD	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	TDI	TDO	TMS/SWDIO	TCK/SWCLK	Hi-Z	Hi-Z	Hi-Z	Hi-Z
00001b	AGT	—	—	—	—	AGTO1	AGTO1	—	—	—	—	—	—	AGTEE1	—	—	—
00010b	GPT*2	—	—	—	—	GTIW	GTIV	GTIU	—	GTOVLO	GTOVUP	GTOULO	GTOUUP	GTETRGD	GTETRGC	—	—
00011b	GPT*2	—	—	GTIOC5B	GTIOC5A	GTIOC4B	GTIOC4A	GTCPPO0	GTCPPO3	GTIOC1B	GTIOC1A	GTIOC0B	GTIOC0A	GTIOC0B	GTIOC0A	—	—
00100b	SCI*1	—	—	CTS_RTS4_A/SS4_A	RXD4_A/SCL4_A/MISO4_A	SCK4_A	TXD4_A/SDA4_A/MOSI4_A	—	—	—	—	—	—	—	—	—	—
00101b	SCI*1	—	—	—	—	—	—	—	—	RXD9_B/SCL9_B/MISO9_B	TXD9_B/SDA9_B/MOSI9_B	CTS_RTS9_B/SS9_B	SCK9_B	RXD1_C/SCL1_C/MISO1_C	TXD1_C/SDA1_C/MOSI1_C	—	—
00110b	SPI*1	—	—	MOSIA_A	RSPCKA_A	SSLA0_A	SSLA1_A	—	—	—	—	—	—	—	—	—	—

Table 21.9 Register settings for input/output pin function (PORT2) (2 of 2)

PSEL[4:0] settings	Function	Pin P200*4	P201	P202	P203	P204	P205	P206	P207	P208	P209	P210	P211	P212	P213	P214	P215
00111b	IIC/ I3C *1	—	—	—	—	SDA1_B	SCL1_B	—	—	—	—	—	—	—	—	—	—
01001b	RTC/ CLKOUT/ ACMPHS/ ETHPHY CLK	—	—	—	—	—	CLKOUT	CLKOUT	—	VCOUT	CLKOUT	—	—	—	—	—	—
01010b	CAC/ ADC16H	—	—	—	—	CACREF	—	—	—	—	—	—	—	—	ADTRG1	—	—
01011b	BUS	—	—	—	—	—	—	CS7	—	—	—	—	—	—	—	—	—
01101b	SCI	—	—	DE4	—	DE4	—	—	—	—	—	—	—	—	—	—	—
01110b	SCI	—	—	—	—	—	—	—	—	—	—	DE9	DE9	—	—	—	—
10000b	CANFD	—	—	CRX0	CTX0	—	—	—	—	CRX1	CTX1	—	—	—	—	—	—
10010b	SSIE*1	—	—	—	—	—	—	SSIDATA1_A	—	—	—	—	—	—	—	—	—
10011b	USBFS	—	—	USB_EXICEN	USB_VBUSEN	USB_OVRCURB	USB_OVRCURA	USB_VBUSEN	—	—	—	—	—	—	—	—	—
10101b	SDHI*1	—	—	SD1CMD_A	SD1CLK_A	SD1WP	SD1CD	SD0DAT7_C	—	—	—	—	—	—	—	—	—
10110b	ESWM (GMII/MII)	—	—	—	—	—	—	ET1_RX_DV	ET1_RXD5	—	—	—	—	—	—	—	—
10111b	ESWM (RMII)	—	—	—	—	—	—	RMII1_CRS_DV	—	—	—	—	—	—	—	—	—
11000b	ESWM (RGMII)	—	—	—	—	—	—	RGMII1_RX_CTL	—	—	—	—	—	—	—	—	—
11001b	GLCDC	—	—	—	—	—	—	—	LCD_DATA9_B	—	—	—	—	—	—	—	—
01111b	CEU	—	—	—	—	—	—	VIO_D0	—	—	—	—	—	—	—	—	—
11110b	ULPT	—	—	ULPTOB1	ULPTOA1	—	—	ULPTOB1	—	—	—	—	—	—	ULPTEE0	—	—
Don't care	Trace (Debug)/ Clock	—	—	—	—	—	—	—	—	—	SWO*5	—	—	EXTAL	XTAL	XCOUT	XCIN/ EXCIN
ASEL bit		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
ISEL bit		NMI	IRQ4	IRQ3-DS	IRQ2-DS	IRQ26	IRQ1-DS	IRQ0-DS	IRQ25	IRQ3	IRQ25	IRQ24	IRQ23	IRQ3	IRQ2	IRQ21	IRQ20
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	—	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	—	—
DSCR[1:0] bit	Drive capability control	—	L*3	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	—	—
289, 303 pins product		✓	✓	—	—	—	—	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
289, 303 pins w/o MIPI product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		✓	✓	—	—	—	—	✓	—	✓	✓	✓	✓	✓	✓	✓	✓
224 pins w/o MIPI product		✓	✓	✓	✓	✓	✓	✓	—	✓	✓	✓	✓	✓	✓	✓	✓

✓: Available
—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance "_A", "_B", or "_C" to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.
Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK}).
Note 3. The driver strength of this port is controlled by PmnPFS.DSCR[1:0] bits. Set the PmnPFS.DSCR[1:0] bits to Low Drive (00b).
Note 4. When using NMI pin interrupt, Port related registers setting are not required.
Note 5. To use this pin function, select the CoreSight TPIU.SPPR.PProtocol bits to SWO and set the TRPORTCR.OE bit to 1. The drive strength of this pin function is controlled by TRPORTCR.DRV[1:0] bits.

Table 21.10 Register settings for input/output pin function (PORT3) (1 of 2)

PSEL[4:0] settings	Function	Pin P300	P301	P302	P303	P304	P305	P306	P307	P308	P309	P310	P311	P312	P313	P314	P315
00000b (value after reset)	Hi-Z/ JTAG/ SWD	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
00001b	AGT	—	AGTO0	—	—	—	—	—	—	—	—	AGTEE1	AGTOB1	AGTOA1	—	—	—
00010b	GPT*2	—	GTOULO	GTOUUP	—	GTOVLO	GTOVUP	GTIW	GTIV	GTIU	—	—	GTADSM1	GTADSM0	—	—	—
00011b	GPT*2	GTIOC3A	GTIOC4B	GTIOC4A	GTIOC7B	GTIOC7A	GTCPPO12	GTCPPO11	GTCPPO10	GTCPPO9	GTCPPO8	GTCPPO7	GTCPPO6	GTCPPO5	—	—	—
00100b	SCI*1	SCK0_A	TXD6_B/ SDA6_B/ MOSI6_B	RXD6_B/ SCL6_B/ MISO6_B	SCK6_B	—	—	—	CTS_RTS6_A/ SS6_A	—	—	—	—	—	—	—	—
00101b	SCI*1	—	—	—	—	—	—	—	—	CTS3_B	CTS9_B	TXD3_B/ SDA3_B/ MOSI3_B	SCK3_B	CTS_RTS3_B/ SS3_B	TXD3_C/ SDA3_C/ MOSI3_C	RXD3_C/ SCL3_C/ MISO3_C	SCK3_C
00110b	SPI*1	SSLA3_B	—	—	—	—	—	—	—	—	—	—	—	—	MISOA_A	SSLA2_A	SSLA3_A
01001b	RTC/ CLKOUT/ ACMPHS/ ETHPHYCLK	—	—	—	—	—	—	—	—	ETHPHYCLK	VCOUT	—	—	—	—	—	—
01010b	CAC/ ADC16H	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ADTRG0	—
01011b	BUS	D2/ DQ2	D1/ DQ1	D0/ DQ0	—	—	—	—	—	—	—	—	—	—	—	—	—
01101b	SCI	DE0	—	—	DE6	—	—	—	DE6	—	—	—	—	—	—	—	—
01110b	SCI	—	—	—	—	—	—	—	—	—	—	—	DE3	DE3	—	—	DE3
10000b	CANFD	—	—	—	—	—	—	—	—	—	—	—	CRX0	CTX0	—	—	—
10011b	USBFS	—	—	—	—	—	—	—	—	—	—	—	—	—	USB_ID	—	—
10101b	SDHI*1	SD0DAT3_B	SD0DAT2_B	SD0DAT1_B	—	SD0DAT0_B	SD0WP	SD0CD	SD0CMD_B	SD0CLK_B	—	—	—	—	SD1DAT0_A	SD1DAT1_A	—
10110b	ESWM (GMII/MII)	—	—	—	—	ET1_TXD3	ET1_TXD2	ET1_TXD1	ET1_TXD0	ET1_TX_ER	ET1_GTX_CLK	ET1_TX_EN	ET1_TX_CLK	ET1_RX_ER	—	—	—
10111b	ESWM (RMII)	—	—	—	—	—	—	RMII1_TXD1	RMII1_TXD0	—	—	RMII1_TX_EN	—	RMII1_RX_ER	—	—	—
11000b	ESWM (RGMII)	—	—	—	—	RGMII1_TXD3	RGMII1_TXD2	RGMII1_TXD1	RGMII1_TXD0	—	RGMII1_TXC	RGMII1_TX_CTL	—	—	—	—	—
11001b	GLCDC	LCD_DATA11_A	LCD_DATA12_A	LCD_DATA13_A	LCD_DATA14_A	—	—	—	—	—	LCD_DATA15_A	LCD_DATA16_A	LCD_DATA17_A	LCD_DATA18_A	—	—	—
11011b	PDMIF	—	—	—	—	—	—	—	—	—	—	—	—	PDMDAT2	—	—	—
01111b	CEU	—	—	—	—	VIO_D15	VIO_D14	VIO_D13	VIO_D12	VIO_D11	VIO_D10	VIO_D9	VIO_D8	VIO_D7	—	—	—
11110b	ULPT	ULPTEVI0-DS	ULPTEE0-DS	ULPTO0-DS	—	ULPTO1	ULPTEE1	ULPTEVI1	ULPTOA1	ULPTOB1	—	—	—	—	—	—	—
Don't care	Trace (Debug)/ Clock	—	—	—	—	TDATA3*3	TDATA2*3	TDATA1*3	TDATA0*3	TCLK*3	—	—	—	—	—	—	—
ASEL bit		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
ISEL bit		IRQ4	IRQ6	IRQ5	IRQ29-DS	IRQ9	IRQ8	IRQ28-DS	IRQ27-DS	IRQ26-DS	IRQ25-DS	IRQ24-DS	IRQ23-DS	IRQ22-DS	IRQ27	IRQ28	IRQ29
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
DSCR[1:0] bit	Drive capability control	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH
289, 303 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	—	—	—	—

Table 21.10 Register settings for input/output pin function (PORT3) (2 of 2)

PSEL[4:0] settings	Function	P300	P301	P302	P303	P304	P305	P306	P307	P308	P309	P310	P311	P312	P313	P314	P315
289, 303 pins w/o MIPI product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		✓	✓	✓	—	✓	✓	✓	✓	✓	✓	✓	✓	—	—	—	—
224 pins w/o MIPI product		✓	✓	✓	—	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

✓: Available
—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance "_A", "_B", or "_C" to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK}).

Note 3. To use this pin function, select the CoreSight TPIU.SPPR.PProtocol bits to parallel port and set the TRPORTCR.OE bit to 1. The drive strength of this pin function is controlled by TRPORTCR.DRV[1:0] bits. How many pins of Trace Data are used can be controlled by TRPORTSZ.PORTSIZE[31:0] bits.

Table 21.11 Register settings for input/output pin function (PORT4) (1 of 2)

PSEL[4:0] settings	Function	P400	P401	P402	P403	P404	P405	P406	P407	P408	P409	P410	P411	P412	P413	P414	P415
00000b (value after reset)	Hi-Z/JTAG/SWD	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
00001b	AGT	AGTIO1	—	—	—	—	AGTIO1	—	AGTIO0	—	—	AGTOB1	AGTOA1	AGTEE1	—	—	—
00010b	GPT*2	—	GTETRGA	—	—	—	—	—	—	GTOWLO	GTOWUP	GTOVLO	GTOVUP	GTOULO	GTOUUP	—	—
00011b	GPT*2	GTIOC6A	GTIOC6B	—	GTIOC3A	GTIOC3B	GTIOC1A	GTIOC1B	GTIOC10B	GTIOC10A	—	GTIOC9B	GTIOC9A	GTCPPO8	GTCPPO7	GTIOC0B	GTIOC0A
00100b	SCI*1	—	—	—	—	—	SCK2_B	TXD2_B/SDA2_B/MOSI2_B	—	—	—	—	—	—	—	RXD4_B/SCL4_B/MISO4_B	TXD4_B/SDA4_B/MOSI4_B
00101b	SCI*1	TXD1_A/SDA1_A/MOSI1_A	RXD1_A/SCL1_A/MISO1_A	SCK1_A	CTS_RTS1_A/SS1_A	CTS1_A	—	—	SCK1_C	RXD3_A/SCL3_A/MISO3_A	TXD3_A/SDA3_A/MOSI3_A	SCK3_A	CTS_RTS3_A/SS3_A	CTS3_A	—	—	—
00110b	SPI*1	—	—	—	—	—	—	SSLA3_C	—	—	—	—	—	—	—	SSLB0_B	RSPCKB_B
00111b	IIC/I3C*1	I3C_SCL0	I3C_SDA0	—	—	—	—	—	SDA0_B	SCL0_B	SDA0_A	SCL0_A	—	—	—	—	—
01001b	RTC/CLKOUT/ACMPHS/ETHPHYCLK	—	—	—	—	—	—	—	RTCOUT	—	—	—	—	—	—	—	—
01010b	CAC/ADC16H	ADTRG1	—	CACREF	AD0FLAG1	AD1FLAG1	—	—	ADTRG0	ADSYNC	ADST1	ADST0	CACREF	—	—	—	—
01011b	BUS	—	—	—	—	—	—	—	CS6	A17	A18	A19	A20	A21	A22	A23	WAIT
01101b	SCI	—	—	—	—	—	DE2	—	—	—	—	—	—	—	—	—	—
01110b	SCI	—	—	DE1	DE1	—	—	—	DE1	—	—	DE3	DE3	—	—	—	—
10000b	CANFD	—	CTX0	CRX0	—	—	—	—	—	—	—	—	—	—	—	CRX1	CTX1
10010b	SSIE*1	AUDIO_CLK	—	AUDIO_CLK	SSIBCK0_A	SSILRCK0/SSIFS0_A	SSITXD0_A	SSIRXD0_A	—	—	—	—	—	—	—	—	—
10011b	USBFS	—	—	—	—	—	—	—	USB_VBUS	USB_VBUSEN	USB_OVRCURA-DS	USB_OVRCURB-DS	USB_ID	USB_EXICEN	—	—	—
10100b	USBHS	—	—	—	—	—	—	—	USBHS_VBUSEN	USBHS_VBUS	USBHS_OVRCURA	USBHS_OVRCURB	USBHS_ID	USBHS_EXICEN	—	—	—
10101b	SDHI*1	SD1CLK_B	SD1CMD_B	SD1DAT0_B	SD1DAT1_B	SD1DAT2_B	SD1DAT3_B	SD1CD	—	—	—	—	—	—	—	—	—

Table 21.11 Register settings for input/output pin function (PORT4) (2 of 2)

PSEL[4:0] settings	Function	Pin P400	P401	P402	P403	P404	P405	P406	P407	P408	P409	P410	P411	P412	P413	P414	P415
10110b	ESWM (GMII/MII)	—	—	ET0_LINKSTA	ET1_WOL	ET0_WOL	ET0_RX_DV	ET0_RXD3	GPTP_PTPOUT3	GPTP_PTPOUT2	GPTP_CAPTURE0	GPTP_MATCH0	GPTP_PTPOUT1	GPTP_PTPOUT0	ET_TAS_STA3	ET1_MDIO	ET1_MDC
10111b	ESWM (RMII)	—	—	—	—	—	RMII0_CRS_DV	—	—	—	—	—	—	—	—	—	—
11000b	ESWM (RGMII)	—	—	—	—	—	RGMII0_RX_CTL	RGMII0_RXD3	—	—	—	—	—	—	—	—	—
01111b	CEU	VIO_D0	VIO_D1	—	—	—	VIO_D2	VIO_D3	—	—	—	—	—	—	—	VIO_CLK	VIO_HD
11110b	ULPT	—	—	—	—	—	—	—	—	ULPTOB0	ULPTOA0	—	—	—	ULPTEE1	—	—
11111b	MIPI	—	—	—	—	—	—	—	—	—	—	—	DSI_TE	—	—	—	—
Don't care	RTC	—	—	RTCIC0*3	RTCIC1*3	RTCIC2*3	—	—	—	—	—	—	—	—	—	—	—
ASEL bit		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
ISEL bit		IRQ0	IRQ5-DS	IRQ4-DS	IRQ14-DS	IRQ15-DS	IRQ30	IRQ31	IRQ22	IRQ7	IRQ6	IRQ5	IRQ4	IRQ20-DS	IRQ18	IRQ9	IRQ8
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
DSCR[1:0] bit	Drive capability control	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH
289, 303 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
289, 303 pins w/o MIPI product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		✓	✓	✓	—	✓	✓	✓	✓	✓	✓	✓	✓	—	—	✓	✓
224 pins w/o MIPI product		✓	✓	✓	—	✓	✓	✓	✓	✓	✓	✓	✓	—	—	✓	✓

✓: Available
—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance "_A", "_B", or "_C" to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK}).

Note 3. To use this pin function, set the associated pin as a general input (set the PmnPFS.PDR and PmnPFS.PMR bits to 0).

Table 21.12 Register settings for input/output pin function (PORT5) (1 of 2)

PSEL[4:0] settings	Function	Pin P500	P501	P502	P503	P504	P505	P506	P507	P508	P509	P510	P511	P512	P513	P514	P515
00000b (value after reset)	Hi-Z/JTAG/SWD	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
00010b	GPT*2	—	—	—	GTOUUP	GTOULO	GTOWUP	GTOWLO	GTADSM0	GTADSM1	—	—	—	—	—	—	—
00011b	GPT*2	GTIOC11B	GTIOC12A	GTIOC12B	GTCPPO6	GTCPPO1	GTCPPO2	GTCPPO3	GTIOC0A	GTIOC0B	GTIOC1A	GTIOC1B	GTIOC0B	GTIOC0A	GTIOC13B	GTIOC13B	GTIOC13A
00100b	SCI*1	RXD8_B/SCL8_B/MISO8_B	TXD8_B/SDA8_B/MOSI8_B	SCK8_B	—	—	—	—	—	—	—	—	CTS_RTS8_A/SS8_A	CTS8_A	SCK8_A	SCK4_C	CTS_RTS4_C/SS4_C
00101b	SCI*1	—	—	—	—	—	—	—	CTS_RTS7_A/SS7_A	CTS5_A	CTS_RTS5_A/SS5_A	RXD5_A/SCL5_A/MISO5_A	—	—	—	—	—
00111b	IIC/I3C*1	—	—	—	—	—	—	—	—	—	—	—	SDA1_A	SCL1_A	—	SDA2_B	SCL2_B
01010b	CAC/ADC16H	CACREF	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
01011b	BUS	—	—	—	A6	A7	A8	A9	A10	A11	A12	A13	—	—	—	—	—

Table 21.12 Register settings for input/output pin function (PORT5) (2 of 2)

PSEL[4:0] settings	Function	Pin															
		P500	P501	P502	P503	P504	P505	P506	P507	P508	P509	P510	P511	P512	P513	P514	P515
01101b	SCI	—	—	DE8	—	—	—	—	—	—	—	—	DE8	—	DE8	DE4	DE4
01110b	SCI	—	—	—	—	—	—	—	DE7	—	DE5	—	—	—	—	—	—
10000b	CANFD	—	—	—	—	—	—	—	—	—	—	—	CRX1	CTX1	—	—	—
10011b	USBFS	USB_VBUSEN	USB_OVRCURA	USB_OVRCURB	—	—	—	—	—	—	—	—	—	—	—	—	—
10101b	SDHI*1	SD1DAT5_A	SD1DAT6_A	SD1DAT7_A	SD0CD	SD0WP	SD0CLK_A	SD0CMD_A	—	—	—	—	—	—	—	—	—
10110b	ESWM (GMII/MII)	—	—	—	—	—	—	—	ET_TAS_STA0	ET_TAS_STA1	ET_TAS_STA2	ET_TAS_STA3	ET1_LINKSTA	ET1_INT	ET0_INT	ET_TAS_STA1	ET_TAS_STA0
11001b	GLCDC	—	—	—	—	—	—	—	LCD_EXTCLK_A	—	—	—	—	—	LCD_TCON3_B	LCD_EXTCLK_B	LCD_CLK_B
11011b	PDMIF	PDMDAT0	PDMDAT1	PDMDAT2	—	—	—	—	—	—	—	—	—	—	—	—	—
01111b	CEU	—	—	—	VIO_D4	VIO_D3	VIO_D2	VIO_D1	VIO_D0	VIO_VD	VIO_HD	VIO_CLK	—	—	VIO_FLD	—	—
11110b	ULPT	—	—	—	—	—	—	—	—	—	ULPTEVI1	ULPTEVI0	—	—	—	—	—
ASEL bit		AN021	AN020	AN019	—	—	—	—	—	—	—	—	—	—	AN016/IVCMP0	—	—
ISEL bit		IRQ24	IRQ25	IRQ26	IRQ6	IRQ7	IRQ8	IRQ9	IRQ10	IRQ1	IRQ2	IRQ3	IRQ15	IRQ14	IRQ31	IRQ13	IRQ12
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
DSCR[1:0] bit	Drive capability control	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH
289, 303 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
289, 303 pins w/o MIPI product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		—	—	—	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	—	—	—
224 pins w/o MIPI product		—	—	—	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	—	—	—

✓: Available
—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance "_A", "_B", or "_C" to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK}).

Table 21.13 Register settings for input/output pin function (PORT6) (1 of 2)

PSEL[4:0] settings	Function	Pin															
		P600	P601	P602	P603	P604	P605	P606	P607	P608	P609	P610	P611	P612	P613	P614	P615
00000b (value after reset)	Hi-Z/JTAG/SWD	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
00001b	AGT	—	—	—	—	—	—	—	—	—	—	—	—	—	AGTO1	AGTO0	—
00010b	GPT*2	—	—	—	—	—	—	—	—	GTOWUP	GTIU	GTOUUP	GTOULO	—	GTETRGA	GTETRGB	GTETRGC
00011b	GPT*2	GTIOC6B	GTIOC6A	GTIOC7B	GTIOC7A	GTIOC8B	GTIOC8A	—	—	GTCPPO4	GTIOC5B	GTIOC4A	GTIOC4B	GTIOC9A	GTIOC9B	GTCPPO9	GTCPPO10
00100b	SCI*1	—	SCK0_B	RXD0_B/SCL0_B/MISO0_B	TXD0_B/SDA0_B/MOSI0_B	CTS_RTS0_B/SS0_B	CTS0_B	—	—	—	TXD0_C/SDA0_C/MOSI0_C	RXD0_C/SCL0_C/MISO0_C	SCK0_C	CTS_RTS0_C/SS0_C	CTS0_C	—	—
00101b	SCI*1	—	—	—	—	—	—	—	—	TXD5_A/SDA5_A/MOSI5_A	—	—	—	—	—	RXD7_A/SCL7_A/MISO7_A	TXD7_A/SDA7_A/MOSI7_A

Table 21.13 Register settings for input/output pin function (PORT6) (2 of 2)

PSEL[4:0] settings	Function	Pin P600	P601	P602	P603	P604	P605	P606	P607	P608	P609	P610	P611	P612	P613	P614	P615
00110b	SPI*1	—	—	—	—	—	—	—	—	—	MISOA_B	RSPCKA_B	MOSIA_B	SSLA0_B	—	—	—
01001b	RTC/CLKOUT/ACMPHS/ETHPHYCLK	—	RTCOUT	—	—	—	—	—	—	—	—	—	CLKOUT	—	—	—	—
01010b	CAC/ADC16H	CACREF	—	—	—	—	—	—	—	CACREF	AD1FLAG1	—	CACREF	—	—	—	—
01011b	BUS	—	—	—	—	—	—	WR3/BC3	D31/DQ31	A14	D7/DQ7	D12/DQ12	D13/DQ13	D14/DQ14	D15/DQ15	WR/WR0/DQM0	WR2/BC2/DQM2
01101b	SCI	—	DE0	—	—	DE0	—	—	—	—	—	—	DE0	DE0	—	—	—
10000b	CANFD	—	—	—	—	—	—	—	—	—	CTX1	CRX1	—	—	—	—	—
10100b	USBHS	—	—	—	—	—	—	—	—	—	—	—	USBHS_VBUSEN	USBHS_OVRCURA	USBHS_OVRCURB	USBHS_ID	USBHS_EXICEN
10110b	ESWM (GMII/MII)	ET1_WOL	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
11001b	GLCDC	—	—	—	—	—	—	—	—	—	LCD_DATA6_A	LCD_DATA5_A	LCD_DATA4_A	LCD_DATA3_A	LCD_DATA2_A	LCD_DATA1_A	LCD_DATA0_A
11100b	OSPI	OM_0_RSTO1	OM_0_WP1	OM_1_SCLKN	OM_1_SCLK	OM_1_SIO7	OM_1_SIO1	OM_1_SIO6	OM_1_DQS	—	—	—	—	—	—	—	—
01111b	CEU	—	—	—	—	—	—	—	—	VIO_FLD	—	—	—	—	—	—	—
11110b	ULPT	ULPTEVI1-DS	ULPTEVI0	ULPTEE0	ULPTO0	—	—	—	—	—	ULPTOA1-DS	ULPTOB1-DS	—	—	—	—	—
ASEL bit		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
ISEL bit		IRQ30	IRQ29	IRQ28	IRQ27	IRQ26	IRQ25	IR24	IRQ23	IRQ22	IRQ29	IRQ16	IRQ17	IRQ18	IRQ19	IRQ20	IRQ7
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
DSCR[1:0] bit	Drive capability control	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH
289, 303 pins product*3		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
289, 303 pins w/o MIPI product*3		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		✓	✓	✓	✓	✓	—	—	—	✓	✓	✓	✓	✓	✓	✓	✓
224 pins w/o MIPI product		✓	✓	✓	✓	✓	—	—	—	✓	✓	✓	✓	✓	✓	✓	✓

✓: Available
—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance “_A”, “_B”, or “_C” to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK}).

Note 3. For BGA303 package, P603 to P607 are connected to the SiP flash inside the package. To access SiP Flash memory, set OSPI in PSEL settings.

Table 21.14 Register settings for input/output pin function (PORT7) (1 of 3)

PSEL[4:0] settings	Function	Pin P700	P701	P702	P703	P704	P705	P706	P707	P708	P709	P710	P711	P712	P713	P714	P715
00000b (value after reset)	Hi-Z/JTAG/SWD	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
00001b	AGT	—	—	—	AGTO1	AGTO0	AGTIO0	AGTIO0	—	—	—	—	AGTEE0	AGTOB0	AGTOA0	—	—
00010b	GPT*2	—	—	—	—	GTADSM0	GTADSM1	—	—	—	—	—	—	—	—	—	—
00011b	GPT*2	GTIOC5A	GTIOC5B	GTIOC6A	GTIOC6B	GTCPPO0	GTCPPO1	GTCPPO2	GTCPPO3	GTCPPO6	GTCPPO5	GTIOC11B	GTIOC11A	GTIOC2B	GTIOC2A	GTIOC12B	GTIOC12A

Table 21.14 Register settings for input/output pin function (PORT7) (2 of 3)

PSEL[4:0] settings	Function	Pin P700	P701	P702	P703	P704	P705	P706	P707	P708	P709	P710	P711	P712	P713	P714	P715
00100b	SCI*1	RXD2_B/SCL2_B/MISO2_B	CTS_RTS2_B/SS2_B	CTS2_B	—	—	—	—	—	SCK4_B	CTS_RTS4_B/SS4_B	CTS4_B	—	—	CTS4_C	TXD4_C/SDA4_C/MOSI4_C	RXD4_C/SCL4_C/MISO4_C
00101b	SCI*1	—	—	—	—	—	CTS1_B	RXD1_B/SCL1_B//MISO1_B	TXD1_B/SDA1_B/MOSI1_B	—	—	—	CTS_RTS1_C/SS1_C	CTS1_C	—	—	—
00110b	SPI*1	MISOA_C	MOSIA_C	RSPCKA_C	SSLA0_C	SSLA1_C	SSLA2_C	—	—	MOSIB_B	MISOB_B	SSLB3_B	SSLB2_B	SSLB1_B	—	—	—
00111b	IIC/I3C *1	—	—	—	—	—	—	—	—	SDA2_A	SCL2_A	—	—	—	—	—	—
01001b	RTC/CLKOUT/ACMPHS/ETHPHYCLK	—	—	—	VCOUT	—	—	ETHPHYCLK	ETHPHYCLK	—	—	—	—	—	—	—	—
01010b	CAC/ADC16H	—	—	—	—	—	—	—	—	CACREF	—	—	—	—	—	—	—
01011b	BUS	—	—	—	—	—	—	—	—	WR1/BC1	CS4	CS5	—	—	—	—	—
01101b	SCI	—	DE2	—	—	—	—	—	—	DE4	DE4	—	—	—	—	—	—
01110b	SCI	—	—	—	—	—	—	—	—	—	—	—	DE1	—	—	—	—
10000b	CANFD	—	—	—	—	CTX0	CRX0	—	—	—	—	—	—	—	—	—	—
10010b	SSIE*1	SSIDATA1_B	SSILRCK1/SSIFS1_B	SSIBCK1_B	—	—	—	—	—	AUDIO_CLK	—	—	—	—	—	—	—
10100b	USBHS	—	—	—	—	—	—	USBHS_OVRCURB-DS	USBHS_OVRCURA-DS	—	—	—	—	—	—	—	—
10101b	SDHI*1	SD1WP	SD1DAT4_B	SD1DAT5_B	SD1DAT6_B	SD1DAT7_B	—	—	—	—	—	—	—	—	—	—	—
10110b	ESWM (GMII/MII)	ET0_RXD2	ET0_RXD1	ET0_RXD0	ET0_RX_CLK	ET0_RX_ER	ET0_TX_EN	ET0_GTX_CLK	ET0_TX_ER	ET0_MDC	ET0_MDIO	ET0_LINKSTA	GPTP_PPS0	GPTP_CAPTURE1	GPTP_MATCH1	GPTP_PPS1	ET_TAS_STA2
10111b	ESWM (RMII)	—	RMII0_RXD1	RMII0_RXD0	RMII0_REF50CK	RMII0_RX_ER	RMII0_TX_EN	—	—	—	—	—	—	—	—	—	—
11000b	ESWM (RGMII)	RGMII0_RXD2	RGMII0_RXD1	RGMII0_RXD0	RGMII0_RXC	—	RGMII0_TX_CTL	RGMII0_TXC	—	—	—	—	—	—	—	—	—
11001b	GLCDC	—	—	—	—	—	—	—	LCD_DATA18_B	—	—	LCD_EXTCLK_B	LCD_DATA19_B	LCD_DATA20_B	LCD_DATA21_B	LCD_DATA22_B	LCD_DATA23_B
11011b	PDMIF	—	—	—	PDMCLK0	PDMCLK1	PDMCLK2	PDMDAT0	PDMDAT1	—	—	—	—	—	—	—	—
01111b	CEU	VIO_D4	VIO_D5	VIO_D6	VIO_D7	VIO_D8	VIO_D9	VIO_D10	VIO_D11	VIO_VD	VIO_D13	VIO_D12	—	—	—	—	—
11110b	ULPT	—	ULPTO1	ULPTO0	—	—	—	—	—	—	—	—	—	—	—	—	—
11111b	MIPI	—	—	—	—	—	—	—	—	—	—	—	—	—	—	DSI_TE	—
ASEL bit		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
ISEL bit		IRQ16-DS	IRQ17-DS	IRQ18-DS	IRQ19-DS	IRQ26	IRQ19	IRQ7	IRQ8	IRQ11	IRQ10	IRQ17	IRQ3	IRQ2	IRQ14	IRQ13	IRQ12
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
DSCR[1:0] bit	Drive capability control	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH
289, 303 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Table 21.14 Register settings for input/output pin function (PORT7) (3 of 3)

PSEL[4:0] settings	Function	Pin P700	P701	P702	P703	P704	P705	P706	P707	P708	P709	P710	P711	P712	P713	P714	P715
289, 303 pins w/o MIPI product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		✓	✓	✓	✓	✓	—	—	—	✓	✓	✓	✓	✓	✓	✓	—
224 pins w/o MIPI product		✓	✓	✓	✓	✓	—	—	—	✓	✓	✓	✓	✓	✓	✓	—

✓: Available
—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance "_A", "_B", or "_C" to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK}).

Table 21.15 Register settings for input/output pin function (PORT8) (1 of 2)

PSEL[4:0] settings	Function	Pin P800	P801	P802	P803	P804	P805	P806	P807	P808	P809	P810	P811	P812	P813	P814	P815
00000b (value after reset)	Hi-Z/ JTAG/ SWD	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
00001b	AGT	AGTOA0	AGTOB0	—	—	—	—	—	—	—	—	—	—	—	—	—	—
00010b	GPT*2	GTIU	GTIV	GTIW	GTETRGC	GTETRGD	—	—	—	—	—	—	—	—	GTETRGA	—	—
00011b	GPT*2	GTIOC11A	GTIOC11B	GTIOC12A	GTIOC12B	GTIOC13A	—	—	GTIOC13A	GTIOC13B	—	GTIOC10A	GTIOC10B	GTIOC11A	GTIOC7B	GTIOC8B	GTIOC8A
00100b	SCI*1	CTS2_A	TXD2_A/ SDA2_A/ MOSI2_A	RXD2_A/ SCL2_A/ MISO2_A	SCK2_A	CTS_RTS2_A/ SS2_A	TXD8_A/ SDA8_A/ MOSI8_A	RXD8_A/ SCL8_A/ MISO8_A	—	—	—	—	—	—	—	—	—
00101b	SCI*1	—	—	—	—	—	—	—	—	RXD7_B/ SCL7_B/ MISO7_B	TXD7_B/ SDA7_B/ MOSI7_B	SCK7_B	CTS7_B	CTS_RTS7_B/ SS7_B	SCK7_A	—	—
01011b	BUS	—	—	—	—	—	—	—	—	—	—	—	—	—	SDCS	—	—
01101b	SCI	—	—	—	DE2	DE2	—	—	—	—	—	—	—	—	—	—	—
01110b	SCI	—	—	—	—	—	—	—	—	—	—	DE7	—	DE7	DE7	—	—
10000b	CANFD	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CRX0	CTX0
10011b	USBFS	—	—	—	—	—	—	—	—	—	—	—	USB_ID	USB_EXICEN	—	USB_DP	USB_DM
10101b	SDHI*1	—	—	—	—	—	—	—	—	—	—	SD1DAT2_A	SD1DAT3_A	SD1DAT4_A	—	—	—
10110b	ESWM (GMII/MII)	—	GPTP_PPS1	—	—	—	ET1_MDIO	ET1_MDC	—	—	—	—	—	—	—	—	—
11001b	GLCDC	—	—	—	—	—	LCD_TCON1_B	LCD_TCON0_B	LCD_TCON2_B	—	—	—	—	—	—	—	—
11011b	PDMIF	—	—	—	—	—	—	—	—	—	—	PDMCLK0	PDMCLK1	PDMCLK2	PDMCLK2	—	—
11100b	OSPI	OM_0_SIO5	OM_0_DQS	OM_0_SIO6	OM_0_SIO1	OM_0_SIO7	—	—	—	OM_0_SCLK	OM_0_SCLKN	—	—	—	—	—	—
01111b	CEU	—	—	—	—	—	VIO_D15	VIO_D14	—	—	—	—	—	—	VIO_D13	—	—
11110b	ULPT	—	—	—	—	—	—	—	—	—	—	ULPTOA0	ULPTOB0	—	—	—	—
11111b	MIPI	—	—	—	—	DSI_TE	—	—	—	—	—	—	—	—	—	—	—
Don't care	USBFS	—	—	—	—	—	—	—	—	—	—	—	—	—	—	USB_DP	USB_DM
ASEL bit		—	—	—	—	—	AN017/ IVCMP0	AN018	—	—	—	—	—	AN022	—	—	—
ISEL bit		IRQ11	IRQ12	IRQ18	IRQ19	IRQ14	IRQ30	IRQ0	IRQ11	IRQ15	IRQ20	IRQ21	IRQ22	IRQ23	IRQ15	IRQ16	IRQ15
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Table 21.15 Register settings for input/output pin function (PORT8) (2 of 2)

PSEL[4:0] settings	Function	Pin P800	P801	P802	P803	P804	P805	P806	P807	P808	P809	P810	P811	P812	P813	P814	P815
NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
DSCR[1:0] bit	Drive capability control	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH
289, 303 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
289, 303 pins w/o MIPI product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		✓	✓	✓	✓	✓	—	—	—	✓	✓	—	—	—	✓	✓	✓
224 pins w/o MIPI product		✓	✓	✓	✓	✓	—	—	—	✓	✓	—	—	—	✓	✓	✓

✓: Available
—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance "_A", "_B", or "_C" to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK}).

Table 21.16 Register settings for input/output pin function (PORT9) (1 of 2)

PSEL[4:0] settings	Function	Pin P900	P901	P902	P903	P904	P905	P906	P907	P908	P909	P910	P911	P912	P913	P914	P915
00000b (value after reset)	Hi-Z/JTAG/SWD	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
00001b	AGT	—	AGTIO1	—	—	—	—	—	—	—	—	—	—	—	—	—	—
00010b	GPT*2	GTADSM0	GTADSM1	—	—	—	—	—	—	—	—	—	—	—	—	—	—
00011b	GPT*2	—	—	GTCPPO13	GTIOC11A	GTIOC11B	GTCPPO13	GTIOC13B	GTIOC13A	GTIOC12B	GTIOC12A	GTCPPO12	GTIOC3B	GTIOC3A	GTCPPO11	GTIOC5B	GTIOC5A
00100b	SCI*1	—	—	—	—	—	—	CTS6_A	SCK6_A	TXD6_A/SDA6_A/MOSI6_A	RXD6_A/SCL6_A/MISO6_A	—	—	—	—	CTS_RTS6_B/SS6_B	CTS6_B
00101b	SCI*1	CTS3_C	CTS_RTS3_C/SS3_C	—	—	—	RXD3_B/SCL3_B/MISO3_B	—	—	—	—	—	—	—	—	—	—
01001b	RTC/CLKOUT/ACMPHS/ETHPHYCLK	—	—	ETHPHYCLK	—	—	—	—	—	—	—	—	—	—	CLKOUT	—	—
01010b	CAC/ADC16H	—	—	—	—	—	AD1FLAG1	AD0FLAG1	ADSYNC	ADST1	ADST0	—	—	—	—	—	—
01011b	BUS	—	—	ALE	—	—	—	—	—	—	—	—	—	—	—	—	—
01101b	SCI	—	—	—	—	—	—	—	DE6	—	—	—	—	—	—	DE6	—
01110b	SCI	—	DE3	—	—	—	—	—	—	—	—	—	—	—	—	—	—
10000b	CANFD	—	—	—	—	—	—	—	—	CRX1	CTX1	—	—	—	—	—	—
10010b	SSIE*1	—	—	AUDIO_CLK	—	—	—	SSILRCK1/SSIFS1_A	SSIBCK1_A	—	—	—	—	—	—	—	—
10011b	USBFS	—	—	—	—	—	—	USB_ID	USB_EXICEN	USB_OVRCURB	USB_OVRCURA	—	—	—	—	—	—
10100b	USBHS	—	—	—	—	—	—	—	—	USBHS_ID	USBHS_EXICEN	—	—	—	—	—	—
10110b	ESWM (GMII/MII)	—	—	—	—	ET1_RXD4	ET1_RX_CLK	ET1_RXD0	ET1_RXD1	ET1_RXD2	ET1_RXD3	ET1_TXD4	ET1_TXD5	ET1_TXD6	ET1_TXD7	—	—
10111b	ESWM (RMII)	—	—	—	—	—	RMII1_REF50CK	RMII1_RXD0	RMII1_RXD1	—	—	—	—	—	—	—	—

Table 21.16 Register settings for input/output pin function (PORT9) (2 of 2)

PSEL[4:0] settings	Function	Pin P900	P901	P902	P903	P904	P905	P906	P907	P908	P909	P910	P911	P912	P913	P914	P915
11000b	ESWM (RGMII)	—	—	—	—	—	RGMII1_RXC	RGMII1_RXD0	RGMII1_RXD1	RGMII1_RXD2	RGMII1_RXD3	—	—	—	—	—	—
11001b	GLCDC	—	—	LCD_DATA3_B	LCD_DATA2_B	LCD_DATA8_B	LCD_DATA19_A	LCD_DATA20_A	LCD_DATA21_A	LCD_DATA22_A	LCD_DATA23_A	LCD_DATA4_B	LCD_DATA5_B	LCD_DATA6_B	LCD_DATA7_B	LCD_DATA0_B	LCD_DATA1_B
11011b	PDMIF	—	—	—	—	—	PDMDAT1	PDMDAT0	PDMCLK2	PDMCLK1	PDMCLK0	—	—	—	—	—	—
01111b	CEU	—	—	VIO_D1	—	—	VIO_D6	VIO_D5	VIO_D4	VIO_D3	VIO_D2	—	—	—	—	—	—
11110b	ULPT	—	—	—	—	—	—	ULPTO1	ULPTEE1	ULPTEVI1	ULPTOA1	—	—	—	—	—	—
ASEL bit		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
ISEL bit		IRQ30	IRQ31	IRQ0	IRQ1	IRQ2	IRQ8	IRQ9	IRQ10	IRQ11	IRQ21-DS	IRQ7	IRQ6	IRQ5	IRQ3	IRQ9	IRQ8
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
DSCR[1:0] bit	Drive capability control	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH
289, 303 pins product		—	—	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
289, 303 pins w/o MIPI product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		—	—	✓	—	—	✓	✓	✓	✓	✓	—	—	—	—	—	—
224 pins w/o MIPI product		—	—	✓	—	—	✓	✓	✓	✓	✓	—	—	—	—	—	—

✓: Available
—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance“_A”, “_B”, or “_C” to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK}).

Table 21.17 Register settings for input/output pin function (PORTA) (1 of 2)

PSEL[4:0] settings	Function	Pin PA00	PA01	PA02	PA03	PA04	PA05	PA06	PA07	PA08	PA09	PA10	PA11	PA12	PA13	PA14	PA15
00000b (value after reset)	Hi-Z/ JTAG/ SWD	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
00010b	GPT*2	GTOVLO	GTOVUP	GTIW	GTIV	GTIU	GTETRGD	GTETRGC	GTETRGB	GTETRGD	—	—	GTIV	GTIW	GTOVUP	GTOVLO	—
00011b	GPT*2	GTCPPO7	GTCPPO8	GTCPPO9	GTCPPO10	GTIOC4B	GTIOC4A	GTIOC7B	GTIOC7A	GTCPPO11	GTCPPO12	GTCPPO13	GTIOC6A	GTIOC6B	GTIOC10A	GTIOC10B	GTIOC7A
00100b	SCI*1	—	—	RXD2_C/ SCL2_C/ MISO2_C	TXD2_C/ SDA2_C/ MOSI2_C	SCK2_C	CTS_RTS2_C/ SS2_C	CTS2_C	—	—	—	—	—	—	—	—	—
00101b	SCI*1	CTS_RTS5_B/ SS5_B	CTS5_B	—	—	—	—	—	CTS7_A	RXD5_B/ SCL5_B/ MISO5_B	TXD5_B/ SDA5_B/ MOSI5_B	SCK5_B	SCK9_C	RXD9_C/ SCL9_C/ MISO9_C	CTS_RTS9_C/ SS9_C	TXD9_C/ SDA9_C/ MOSI9_C	CTS9_C
01001b	RTC/ CLKOUT/ ACMPHS/ ETHPHY CLK	—	—	—	—	—	—	—	VCOUT	—	—	—	—	—	—	—	—
01010b	CAC/ ADC16H	AD1FLAG1	AD0FLAG1	ADSYNC	ADST1	ADST0	—	—	—	—	—	—	—	—	—	—	—
01011b	BUS	A5	A4	A3	A2	A1/ DQM3	A0/ BC0/ DQM1	CS1/ CKE	RD	CS0/ WE	CS3/ CAS	CS2/ RAS	D8/ DQ8	D9/ DQ9	D10/ DQ10	D11/ DQ11	EBCLK/ SDCLK
01101b	SCI	—	—	—	—	DE2	DE2	—	—	—	—	—	—	—	—	—	—
01110b	SCI	DE5	—	—	—	—	—	—	—	—	—	DE5	DE9	—	DE9	—	—

Table 21.17 Register settings for input/output pin function (PORTA) (2 of 2)

PSEL[4:0] settings	Function	Pin PA00	PA01	PA02	PA03	PA04	PA05	PA06	PA07	PA08	PA09	PA10	PA11	PA12	PA13	PA14	PA15
10101b	SDHI*1	SD0DAT7_A	SD0DAT6_A	SD0DAT5_A	SD0DAT4_A	SD0DAT3_A	SD0DAT2_A	SD0DAT1_A	SD0DAT0_A	—	—	—	—	—	—	—	—
11001b	GLCDC	LCD_CLK_A	LCD_TCON0_A	—	—	—	—	—	—	LCD_TCON3_A	LCD_TCON2_A	LCD_TCON1_A	—	—	—	—	—
11011b	PDMIF	—	—	—	—	—	PDMDAT2	PDMDAT1	PDMDAT0	—	—	PDMCLK0	—	—	—	—	PDMCLK1
01111b	CEU	VIO_D5	VIO_D6	VIO_D7	VIO_D8	VIO_D9	VIO_D10	VIO_D11	VIO_D12	—	—	VIO_D15	—	—	—	—	VIO_D14
ASEL bit		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
ISEL bit		IRQ22	IRQ21	IRQ31	IRQ20	IRQ19	IRQ18	IRQ17	IRQ16	IRQ6	IRQ5	IRQ4	IRQ10	IRQ11	IRQ12	IRQ13	IRQ14
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
DSCR[1:0] bit	Drive capability control	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH
289, 303 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
289, 303 pins w/o MIPI product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins w/o MIPI product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

✓: Available
—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance "_A", "_B", or "_C" to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK}).

Table 21.18 Register settings for input/output pin function (PORTB) (1 of 2)

PSEL[4:0] settings	Function	Pin PB00	PB01	PB02	PB03	PB04	PB05	PB06	PB07
00000b (value after reset)	Hi-Z/ JTAG/ SWD	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
00011b	GPT*2	GTCPPO4	GTCPPO2	GTCPPO0	GTCPPO1	GTCPPO3	GTCPPO4	GTIOC9A	GTIOC9B
00101b	SCI*1	SCK1_B	CTS_RTS1_B/ SS1_B	RXD5_C/ SCL5_C/ MISO5_C	TXD5_C/ SDA5_C/ MOSI5_C	SCK5_C	CTS5_C	CTS_RTS5_C/ SS5_C	—
01010b	CAC/ ADC16H	ADST0	AD1FLAG1	ADST1	ADSYNC	AD0FLAG1	—	—	—
01011b	BUS	—	ALE	—	—	—	—	—	—
01110b	SCI	DE1	DE1	—	—	DE5	—	DE5	—
10100b	USBHS	USBHS_VBUSEN	—	—	—	—	—	—	—
10110b	ESWM (GMII/MII)	ET0_TXD0	ET0_TX_CLK	ET0_TXD1	ET0_TXD2	ET0_TXD3	ET0_TXD7	ET0_TXD6	ET0_TXD5
10111b	ESWM (RMII)	RMII0_TXD0	—	RMII0_TXD1	—	—	—	—	—
11000b	ESWM (RGMII)	RGMII0_TXD0	—	RGMII0_TXD1	RGMII0_TXD2	RGMII0_TXD3	—	—	—
11001b	GLCDC	LCD_DATA17_B	LCD_DATA13_B	LCD_DATA16_B	LCD_DATA15_B	LCD_DATA14_B	LCD_DATA12_B	LCD_DATA11_B	LCD_DATA10_B
11011b	PDMIF	PDMDAT2	—	—	—	—	—	—	—
01111b	CEU	—	VIO_FLD	VIO_VD	VIO_HD	VIO_CLK	—	—	—
ASEL bit		—	—	—	—	—	—	—	—
ISEL bit		IRQ10	IRQ12	IRQ11	IRQ13	IRQ9	IRQ15	IRQ0	IRQ1
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓
DSCR[1:0] bit	Drive capability control	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH
289, 303 pins product		✓	✓	✓	✓	✓	✓	✓	✓

Table 21.18 Register settings for input/output pin function (PORTB) (2 of 2)

PSEL[4:0] settings	Function	Pin PB00	PB01	PB02	PB03	PB04	PB05	PB06	PB07
289, 303 pins w/o MIPI product		✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		✓	✓	✓	✓	✓	—	—	—
224 pins w/o MIPI product		✓	✓	✓	✓	✓	—	—	—

✓: Available
—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance "_A", "_B", or "_C" to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK}).

Table 21.19 Register settings for input/output pin function (PORTC)

PSEL[4:0] settings	Function	Pin PC00	PC01	PC02	PC03	PC04	PC05	PC06	PC07	PC08	PC09	PC10	PC11	PC12	PC13	PC14	PC15
00000b (value after reset)	Hi-Z/ JTAG/ SWD	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
00010b	GPT*2	—	—	—	—	—	—	—	—	—	—	—	—	—	—	GTADSM1	GTADSM0
00011b	GPT*2	GTCPPO7	GTCPPO6	GTCPPO5	GTCPPO4	GTCPPO3	GTCPPO2	GTCPPO1	GTCPPO0	GTCPPO8	—	GTCPPO13	GTCPPO12	GTCPPO11	GTCPPO10	GTCPPO9	—
00100b	SCI*1	—	—	—	—	—	—	—	—	—	—	—	CTS_RTS6_C/ SS6_C	SCK6_C	RXD6_C/ SCL6_C/ MISO6_C	TXD6_C/ SDA6_C/ MOSI6_C	CTS6_C
00101b	SCI*1	CTS_RTS7_C/ SS7_C	CTS7_C	SCK7_C	TXD7_C/ SDA7_C/ MOSI7_C	RXD7_C/ SCL7_C/ MISO7_C	—	—	—	—	—	—	—	—	—	—	—
01011b	BUS	D30/ DQ30	D29/ DQ29	D28/ DQ28	D27/ DQ27	D26/ DQ26	D25/ DQ25	D24/ DQ24	D23/ DQ23	D22/ DQ22	D21/ DQ21	D20/ DQ20	D19/ DQ19	D18/ DQ18	D17/ DQ17	D16/ DQ16	A16
01101b	SCI	—	—	—	—	—	—	—	—	—	—	—	DE6	DE6	—	—	—
01110b	SCI	DE7	—	DE7	—	—	—	—	—	—	—	—	—	—	—	—	—
10000b	CANFD	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CRX1
10110b	ESWM (GMII/MII)	—	—	—	—	—	—	—	—	—	—	—	ET0_MDC	ET0_MDIO	ET0_INT	ET0_WOL	—
11100b	OSPI	OM_1_SIO5	OM_1_SIO0	OM_1_SIO3	OM_1_SIO4	OM_1_SIO2	OM_1_CS1	OM_1_ECSINT1	OM_1_RESET	OM_1_CS0	OM_1_RSTO1	OM_1_WP1	—	—	—	—	—
ASEL bit		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
ISEL bit		IRQ28	IRQ27	IRQ26	IRQ25	IRQ24	IRQ23	IRQ22	IRQ21	IRQ29	IRQ5	IRQ4	IRQ3	IRQ2	IRQ1	IRQ0	IRQ30
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
DSCR[1:0] bit	Drive capability control	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH
289, 303 pins product*3		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
289, 303 pins w/o MIPI product*3		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		—	—	—	—	—	—	—	—	—	—	—	✓	✓	✓	✓	✓
224 pins w/o MIPI product		—	—	—	—	—	—	—	—	—	—	—	✓	✓	✓	✓	✓

✓: Available
—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance "_A", "_B", or "_C" to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK}).

Note 3. For BGA303 package, PC00 to PC07 are connected to the SiP Flash memory inside the package. To access SiP Flash memory, set OSPI in PSEL settings.

Table 21.20 Register settings for input/output pin function (PORTD)

PSEL[4:0] settings	Function	Pin							
		PD00	PD01	PD02	PD03	PD04	PD05	PD06	PD07
00000b (value after reset)	Hi-Z/ JTAG/ SWD	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
00010b	GPT*2	GTOWLO	—	—	—	—	—	—	—
00011b	GPT*2	GTCPPO5	GTCPPO2	GTCPPO1	GTIOC3B	GTIOC3A	GTIOC2B	GTIOC2A	GTCPPO0
00100b	SCI*1	—	SCK8_C	TXD8_C/ SDA8_C/ MOSI8_C	RXD8_C/ SCL8_C/ MISO8_C	CTS_RTS8_C/ SS8_C	CTS8_C	—	—
00101b	SCI*1	SCK5_A	—	—	—	—	—	—	—
01011b	BUS	A15	—	—	—	—	—	—	—
01101b	SCI	—	DE8	—	—	DE8	—	—	—
01110b	SCI	DE5	—	—	—	—	—	—	—
10000b	CANFD	CTX1	—	—	—	—	—	—	—
10100b	USBHS	—	—	—	USBHS_EXICEN	USBHS_ID	USBHS_OVRCURB	USBHS_OVRCURA	USBHS_VBUSEN
10101b	SDHI*1	—	SD0DAT2_C	SD0DAT1_C	SD0DAT0_C	SD0CMD_C	SD0CLK_C	SD0WP	SD0CD
10110b	ESWM (GMII/MII)	—	ET1_RXD6	ET1_RXD7	ET0_RXD4	ET0_RXD5	ET0_RXD6	ET0_RXD7	ET0_TXD4
ASEL bit		—	—	—	—	—	—	—	—
ISEL bit		IRQ23	IRQ22	IRQ21	IRQ21	IRQ20	IRQ19	IRQ18	IRQ17
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓
DSCR[1:0] bit	Drive capability control	L/M/ H/HH	L/M/ H/HH	L/M/ H/HH	L/M/ H/HH	L/M/ H/HH	L/M/ H/HH	L/M/ H/HH	L/M/ H/HH
289, 303 pins product		✓	✓	✓	✓	✓	✓	✓	✓
289, 303 pins w/o MIPI product		✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		✓	—	—	—	—	—	—	—
224 pins w/o MIPI product		✓	—	—	—	—	—	—	—

✓: Available
—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance “_A”, “_B”, or “_C” to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK}).